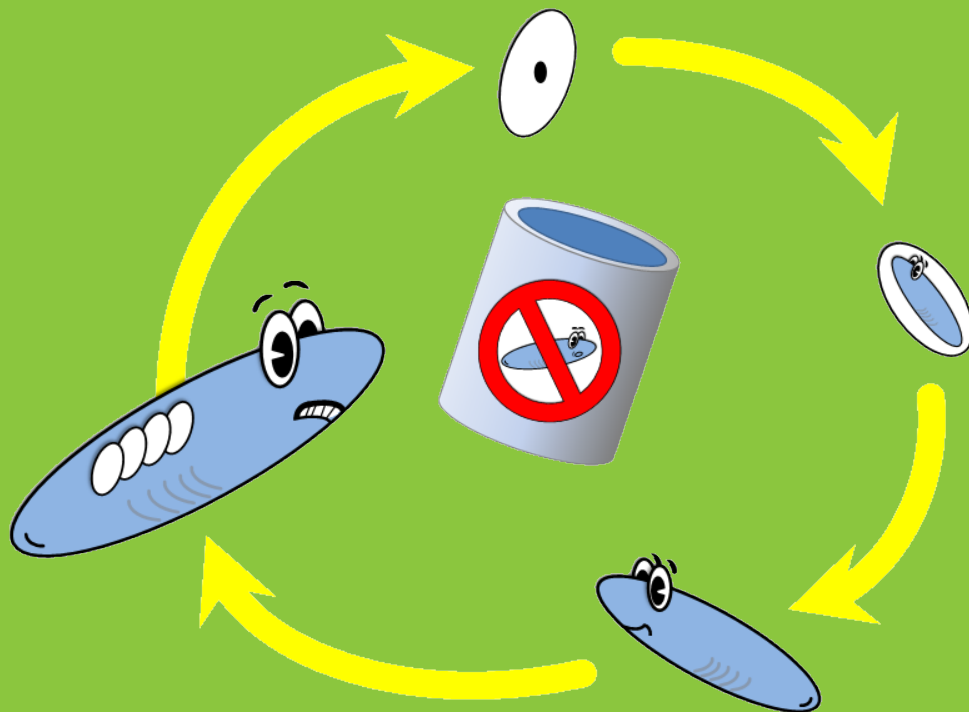


Making Sense of Chemical Stress

APPLICATION OF
DYNAMIC ENERGY BUDGET THEORY
IN ECOTOXICOLOGY AND STRESS ECOLOGY



Tjalling JAGER

Making Sense of Chemical Stress

Application of Dynamic Energy Budget Theory in Ecotoxicology and Stress Ecology

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Version 2.3, Date: June 24, 2025

This e-book is published with Leanpub, and can be downloaded from https://leanpub.com/debtox_book. The book can be obtained totally free of charge. However, please consider paying for it: that gives me the opportunity to continue working on this book, and on other ones. A version log can be found on <http://www.debtox.info/book.html>.

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Alternatively, refer to one of the papers in the open literature. E.g., [127] for the general concepts, [125] for toxicants in the standard DEB animal model, [141] for mixture toxicity in the scaled standard model, [142] for the classical DEBtox model, [132] for the basic DEBkiss model, or [107] for the updated DEBtox model based on DEBkiss.

If you spot errors (spelling, grammar or conceptual), or want to provide other feedback on the book, please contact me by email (see address below).

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Preface

Who should read this book?

Have you ever asked yourself why the effects of toxic chemicals depend on the exposure time? Or asked why stress effects on growth and reproduction are so different in the same organism, even though these responses must be linked in some causal way? Did you ever wish to *understand* toxic effects, so that you can make an educated prediction of effects under other conditions? Or do you want to understand why toxic effects depend on the presence of other factors such as temperature, food density, and life stage? This book offers a framework to address those questions by taking a radically different approach than what is common in ecotoxicology and stress ecology: by simplifying biological reality to an enormous extent. In this book, I will present a ‘mechanistic’ or ‘biology-based’ treatment of chemical effects. The main focus lies on one particularly useful framework for the interpretation of toxic effects, namely Dynamic Energy Budget (DEB) theory, and more specifically, the formulation by Bas Kooijman in 2010 [155] (with a few excursions to the derived, simplified, framework DEBkiss [132, 109]). Even if you are not convinced that this theory is the way to go for your particular problem, knowledge of the concepts behind it allows you to examine your (and other people’s) data and models more critically.

This is not a cookbook with recipes for how to derive *the* toxicity of a chemical from your test data. First and foremost, it is an open invitation to start thinking about toxic effects on organisms as the result of underlying processes; processes in time. We need to treat the individual organism as a system. Furthermore, this book is an invitation to focus on the generalities that link all species and all toxicants, instead of losing ourselves in the details that make them unique. Recognising and understanding the dominant processes governing the toxic response is invaluable for understanding the effects of toxicants in a laboratory test. This understanding, in turn, is crucial to compare effects between species and between chemicals, to discover meaningful patterns, and to make science-based predictions for the real environment, under conditions far removed from those in the laboratory. However, I also want to show you how stress in general (and toxicants in particular) can help to provide insight into the basic structure of metabolic organisation in organisms.

This book covers a lot of fields: biology, (eco)toxicology, chemistry, modelling and statistics. I will not dive into any of these fields in great depth; the message is in their interconnection. There is, as far as I know, no education to properly prepare you for a multi-disciplinary arena such as this. For this reason, I attempted to write this book for a broad audience, assuming no specific background knowledge, and keeping it math-free. However, training in science and in abstract thinking is needed to fully appreciate all of the concepts presented (and some knowledge of ecotoxicology and general biology

would help), and more training would be needed to effectively apply the models.

Why a book?

Simply because a book like this did not exist. Ecotoxicological textbooks do not address the questions I raised in the beginning of this preface, and if they touch upon these subjects, they stick to descriptions. The DEB book of Bas Kooijman [155] has a chapter on toxicant effects. However, that chapter is a tough read as it contains a lot of detail in a highly condensed form. Furthermore, the book presents DEB theory over its full width of application, which will deter many an ecotoxicologist. There exists a dedicated booklet on ‘DEBtox’ [159], presenting a DEB-based analysis for standard toxicity tests: acute survival, juvenile fish growth, *Daphnia* reproduction, and algal population growth. However, it is out of print and is more a collection of scientific papers than a coherent treatise. Furthermore, it presents the equations as such, without paying much attention to explaining the underlying concepts.

Since 2002, I have been working on toxicants in DEB theory, and have tried to explain what I was doing in a considerable number of papers, lectures and courses. I noticed that there is quite a learning curve to DEB theory. Even though the basic concepts of the theory are truly simple, and can be explained in ten minutes, they constitute an extreme abstraction of living systems. Such a level of abstraction is hardly part of the biological scientific tradition, and might lead to feelings of discomfort in the unsuspecting. Furthermore, even though the concepts are simple, following them to their logical consequences is not. Application of the theory almost always requires mathematics, enhancing the feelings of discomfort in many among the audience. Discomfort easily leads to disbelief. In a mathematical model, it is relatively easy to hide a few *ad hoc* factors to get a good fit to a set of data. Of course, all models are simplifications, and thus ‘wrong’, but how can you be sure that a model is actually useful for some purpose?

Models (at least, all useful ones) follow from assumptions. In fact, they should follow *uniquely* from a set of assumptions. Once you accept these assumptions, and given a correct implementation, you should also accept the model and its implications. If you do not trust the model’s predictions, it is wise to start by scrutinising the assumptions. The purpose of this book is thus to clarify the assumptions underlying DEB models for the analysis of toxic effects, with a high level of transparency. Once these assumptions are clear, it will be easier to interpret the model’s fit to actual data sets, and the predictions made from them. For most biologists and ecotoxicologists, math is not helpful to explain something, and probably even a hindrance. To apply DEB models in ecotoxicology, you do not need to be good at math, but you do need a firm grip on the concepts and assumptions.

For the underlying math, I like to refer you to other books and papers. A list with links and/or free downloads is offered at the book’s support page (see below). The standard work on DEB theory is of course the 2010 book by Bas Kooijman [155] (although that book is not up to date when it comes to toxicant effects). For a more general introduction into modelling, and the math and stats needed for TKTD modelling, I advise my e-book on ‘Mechanistic modelling essentials’ [110] (not free but friendly priced). For an in-depth treatise on reserve-less DEBkiss models, including equations and derivations, and including up-to-date modules for toxicant effects, you can consult the dedicated e-book [109] (which can be downloaded for free).

Limitations of this book

To limit the size of this book, and to allow for a more coherent discussion of concepts, I will limit myself to applications involving heterotrophic organisms (mainly animals), and more specifically invertebrates (and, to some extent, fish). The reasons to select this group is that I personally have most experience with them, and the data sets that are routinely collected for these organisms are often directly amenable to a DEB-based treatment. I realise that by limiting myself to a selection of organisms, I neglect one of the most important achievements of DEB theory: the unification of all living organisms into a single, coherent, quantitative theory. Certainly, there are very good examples of DEB application to stressor effects in other groups of organisms such as toxicity in algae [164, 41], and birds [249] and mammals [199]. However, a treatment of these developments might distract from the general message that I want to convey.

Support on the web

The supporting website for this book is <http://www.debtox.info>. Here, you will find software (as toolboxes for Matlab) to perform the calculations, lists of publications that apply these concepts, and information on courses. This site also contains the dedicated support page for this e-book (<http://www.debtox.info/book.html>), which hosts/links documents that presents the mathematical formulations, their derivations, as well as alternative formulations to explore. In addition, I maintain a version log on this web page to keep track of the development of the book.

For more DEB-related information, check out the website of the department of Theoretical Biology: <http://www.bio.vu.nl/thb/deb>. Even though the department no longer exists, the website is still there. More up-to-date information can be obtained from the DEB portal: <https://debportal.debtheory.org/docs/>.

What's that thing on the cover?

The creature on the cover, and in several figures in the book, is a PHylogenetically Indeterminate Life form, or 'Phil' for short. Phil is inspired by the creature that graces the cover of the third edition of the DEB book [155], and is used to illustrate general principles without focusing on specific species. In fact, a cartoon is a model: a simplification of a complex real system, brought back to its essence. Using a cartoon organism instead of a real one thus fits extremely well with the message I want to convey.

Notes for the update to version 2.0

The update to version 2.0 of the book involved a rather major reshuffling of the text. The conceptual switch that was made was to put 'damage' into a central position (whereas this concept has been largely ignored for a long time in a DEBtox context). This reflects the developments for survival modelling with GUTS [118], which has now also been adopted in DEBkiss [109], and more recently in updates of simplified (DEBtox2019, [107]) and complete (stdDEB-TKTD, [125]) DEB-based TKTD models. Not only is it a good idea to strive for harmonisation between the different modelling frameworks, I also feel that considering damage is essential for TKTD models in general. The 'old' models are still in there, but they form a special case of the overarching general

model. This reshuffling and rewriting mainly affected the old chapters on toxicokinetics and toxicodynamics.

Acknowledgements

Firstly, I am indebted to Bas Kooijman whose work on DEB theory laid the foundation on which this book is built. Furthermore, I would like to thank all of my former colleagues at the department of Theoretical Biology, and the other DEB and TKTD enthusiasts around the world, for their inspiring papers, presentations and discussions. Thanks to L^AT_EX for providing the platform to write this book, and thanks to WikiPedia (<http://www.wikipedia.org>) for many interesting facts.

Warnings

I noticed that some of my e-books are offered through other websites, after removal of the title page. Please download this e-book from Leanpub, which is the only place where it is legitimately offered (even for free). That will ensure that you have a complete and correct version, and that you are notified as soon as a new version is available.

This book *will* inevitably contain errors. If you spot one, please let me know so that I can include corrections in updates of this book. I do not accept liability or responsibility for any damage or costs incurred as a result of these errors.

Tips for reading in a PDF viewer

There is no need to print this book; you can read the PDF in a viewer, such as Acrobat Reader or SumatraPDF. In this way, you can use the hyperlinks in this document (shown in blue) to jump to the corresponding reference, footnote, figure, section, etc. Unfortunately, it seems that most viewers do not have a simple ‘back’ button to return to the point where you were reading before jumping with the hyperlink. I recently found out that, in the desktop versions of both Acrobat Reader and SumatraPDF, you can use the shortcut **alt-left arrow** to jump back. No such luck for the mobile version of Acrobat (yet), as far as I can tell.

Chapter 1

Setting the scene

This is a book about the effects of chemical stress on organisms. It is an attempt to construct a general framework to quantitatively understand, and ultimately predict, the biological effects of chemicals over time. In writing this book, I realised that I need to be more specific about the things I want to discuss. Such a limitation is necessary for me to maintain focus in my discussion (which is difficult enough as it is), and for the reader to understand why it is helpful (in my opinion even inevitable) to work in an energy-budget framework. Different choices in scope would lead (and have already led) to very different books. Even though “effects of chemical stress on organisms” sounds like a well-demarcated research area, I do not think it is.

1.1 Limiting the scope

The world is full of things, and all material things are made of chemicals. Unfortunately, we cannot divide chemicals into toxic and non-toxic ones. Paracelsus (1493-1541) was right on the mark when he wrote: “All things are poison and nothing is without poison, only the dose permits something not to be a poison.” Many chemicals are required by organisms for their normal functioning (nutrients). I will not talk about nutrients explicitly, but restrict the discussion to chemicals that are not part of the organism’s ‘normal’ functioning, or are present in levels exceeding the requirements for such functioning. I realise that this definition of ‘chemical stress’ is a bit vague, but it will have to do for now. Even though the focus lies on chemicals, this book has a lot to offer for researchers interested in non-toxicant environmental stress (e.g., food, temperature, disease or pH stress), because the same principles can often be applied. Most of the time, I will be talking about the effects of a single toxicant in isolation. However, it is good to realise that organisms are always exposed to a mixture of chemicals; even in an experimental toxicity test, inevitably, other chemicals will be present in the test medium (although usually at non-toxic levels). In the real world, mixture exposure is the norm, although experimental testing and risk assessment mainly focus on single chemicals.

There are millions of species of organism, so clearly, I want to restrict myself in the biological scope too. The concepts I present are equally valid for all forms of life on this planet (and likely also on others), but I will only work out the case for animals. More specifically, the focus will be on multi-cellular ectotherms, primarily invertebrates and fish. Even though this group represents only a small fraction of the

total number of species on the planet, they have something special. They are popular species in chemical-stress research, they form a group that is homogeneous enough to be described by the same basic model structure (as I will discuss in Chapter 2), and furthermore, the data sets that are routinely collected for these organisms are often directly amenable to the type of analysis that I will present. The last reason is a personal one: my experience with such critters is greater than the other forms of life, which makes it easier for me to write this book.

Another important set of restrictions is in the organisation levels that I will treat. Chemical effects start at the molecular level, and work their way up to the ecosystem, and even global scale. I will focus on the effects on an individual's life cycle, and thus on life history traits or endpoints¹ such as growth, development, reproduction and survival. The individual level is of key interest as it is possible to work with mass and energy balances, and because individuals are the units of natural selection and the building blocks of populations and ecosystems [106]. I will make some excursions to lower and higher levels of organisation, but the individual will be the basis. This implies that I will not deal (explicitly) with effects at the molecular and tissue level, and not with effects on ecosystems, even though there are clear links with the individual level (in fact, it is the individual level that connects these two worlds). In this book, the focus is on understanding and predicting the effects of chemicals on individual-level traits over time, over the entire life cycle of the individual (in principle, from egg to death).

The final restriction I pose myself is that I want to provide a general framework. That is, not specific for a chemical, species or effect. Making a model that accurately predicts the effects of chemical A on trait B of species C is very nice, but the number of different combinations of A, B and C is quite large. In my opinion, there is a need for generalisation as we cannot ever hope to test all the relevant permutations. The intellectual challenge in this book is to provide a framework that applies to all A, B and C within the restrictions posed above. In the case studies, it will become clear that biology often defies a strict generalisation, and more specific auxiliary assumptions will creep in.

With these restrictions in mind, I hope that the subsequent sections in this chapter, and my observations on current research in fields dealing with chemical stress, can be placed in its proper perspective.

1.2 Many faces of chemicals stress

Chemical stress is not something that humans have introduced on this planet; it is as old as life itself. The earth's mantle contains a range of compounds that can affect organisms negatively (for example metals and sulphur compounds). The appearance of free oxygen in the atmosphere (produced by photosynthesis in bacteria), some 2.4 billion years ago, is thought to have caused a mass extinction among the anaerobic² organisms dominating before that time. Incomplete combustion of organic matter is accompanied by the release of a range of particularly toxic organic chemicals such as polycyclic aromatic hydrocarbons and dioxins. Organisms themselves also produce all kinds of (sometimes very complex) chemical compounds, and put them to cunning use to aid

¹In ecotoxicology, the term 'endpoint' is often used to denote a life-history trait that is observed to see if it responds to toxicant exposure.

²Anaerobic organisms function without the need for oxygen. For many of them, oxygen is in fact deadly.

their own survival. In this section, I will put ‘chemical stress’ in a broad perspective, providing examples of naturally-occurring intoxication, and after that, shortly discuss the role that human development is playing.

Examples of chemical use in nature

Many organisms have evolved chemicals to kill or repel their enemies. A broad variety of plants produce secondary metabolites to deter grazers. The perennial plant pyrethrum (*Chrysanthemum cinerariaefolium*) produces potent insecticides (pyrethrins) with neurotoxic activity (especially toxic to insects), which at lower doses seem to repel insects. The synthetic pyrethroid insecticides are derived from these naturally-produced chemicals. Many plant species produce tannins: bitter-tasting polyphenolic compounds. Consumption of large amounts of the tannin-rich acorns is known to be problematic for cattle. Water hemlock (*Cicuta sp.*) produces cicutoxin,³ a highly toxic unsaturated aliphatic alcohol. This compound acts as a stimulant in the central nervous system, resulting in seizures. Livestock is especially at risk, leading to this plant’s common name ‘cowbane’.



Figure 1.1: The foxglove (*Digitalis purpurea*) contains potent cardiac glycosides, which can easily be fatal for humans. A purified component (digoxin) is used for the treatment of heart conditions. This drawing from “Köhlers Medizinal-Pflanzen”, by Franz Eugen Köhler (1887).

³There is a strong tendency in biology and toxicology to relate the name of a toxicant to the species that produces it. Which makes a lot of sense, of course.

A bit closer to home, in fact, *at home*, many of our house and garden plants pack a powerful punch. For humans, the deadliest common houseplant is probably the oleander (*Nerium oleander*). Its sap contains a glycoside (oleandrin) that causes gastrointestinal and cardiac effects. The foxgloves (*Digitalis sp.*) contain similar glycosides, and are also extremely toxic. However, one of the purified glycosides from the foxglove (digoxin) is used for treatment of heart conditions.⁴ Some more moderately toxic plants in and around the house are dumb cane (*Dieffenbachia sp.*, its English common name refers to the effects of needle-shaped calcium oxalate crystals on the throat when chewing on the leaf), hortensia (*Hydrangea sp.*), Swiss cheese plant (*Monstera deliciosa*, the second part of the scientific name probably refers to the tasty ripe fruit), and most spurges (family Euphorbiaceae). The yews (*Taxus sp.*) that are commonly planted in gardens and parks contain highly poisonous alkaloids (known as taxanes), which are also used in chemotherapy because they inhibit cell division. In fact, alkaloids are very popular defence chemicals among plants; it is estimated that 10-25% of the higher plants produce these compounds.

Not all of these poisons are effective in repelling curious mammals; the alkaloids include some of our favourite drugs such as caffeine, cocaine and nicotine, and pharmaceuticals such as morphine, codeine, ephedrine and quinine. Humans have also discovered that some alkaloids (commonly referred to as curare) can be used efficiently in hunting, to poison arrows and darts. A different use by humans of plant toxins is, oddly enough, in fishing. Several tropical and subtropical plant species (e.g., in the genus *Lonchocarpus* and *Derris*) produce rotenone, a respiratory inhibitor (interfering with the electron transport chain in mitochondria). Rotenone is highly toxic to insects and fish, but only mildly toxic for mammals. Various indigenous tribes use plant extracts in fishing: they collect the dead or dying fish that float to the surface for consumption.

The use of defensive chemicals is of course not restricted to plants. Many fungi produce potent toxins such as aflatoxin (by several *Aspergillus* species). The exact reason why these organisms do so is not so clear, although it may be to protect themselves from being eaten by nematodes, amoeba and springtails. Other fungi produce anti-insect toxins to protect the plants with which they live symbiotically [69]. Also animals use chemicals to protect themselves from being eaten. A famous example are the puffer fish (family Tetraodontidae), many of which produce the extremely potent neurotoxin tetrodotoxin, with no known antidote for humans. Despite (or perhaps thanks to) this toxicity, the well-prepared fugu is considered a delicacy in Japan. Another famous example are the poison dart frogs (family Dendrobatidae) that excrete alkaloid poisons from their skin. Alkaloids are also used by ladybirds (family Coccinellidae) to prevent being eaten. Toads from the genus *Bufo* apply, amongst other components, glycosides, similar to that of the oleander and foxgloves to protect themselves. Some animals are able to re-use the toxins from their food; as an example, the caterpillar of the cinnabar moth (*Tyria jacobaeae*) feeds on ragwort (*Jacobaea vulgaris*) and assimilates the bitter tasting alkaloids to become unpalatable itself. Chemical protection can also take on a more active form such as the apitoxin that honey bees (*Apis sp.*) use in their sting to protect their hive.

Some organisms use chemicals to gain a competitive advantage. When stressed, *Penicillium* fungi produce a compound that causes cell death in bacteria (which led

⁴Which underlines that the difference between a poison and a cure can just be in the dosing.



Figure 1.2: Ragwort (*Jacobaea vulgaris*) contains a range of toxic alkaloids that make it particularly dangerous for horses and cattle (especially dried, as the bitter taste is lost, but not the toxicity). The cinnabar moth (*Tyria jacobaeae*) assimilates the alkaloids to protect itself. Left plate: a reproduction of a painting by the Swedish botanist C. A. M. Lindman (1856 - 1928), taken from his book “Bilder ur Nordens Flora”. Right plate: taken from John Curtis’s “British Entomology” Volume 5 (1840s).

to the first ‘modern’ antibiotic treatment⁵). The black walnut (*Juglans nigra*) secretes a poison (juglone) from its roots that acts as a respiratory inhibitor to some other plant species, complicating gardening in its neighbourhood. Some endoparasites, such as trematodes in snails induce their host to grow to a much larger body size than usual (gigantism). This feat is probably accomplished by some chemical released by the parasite.

A range of predators employ toxic chemicals to kill or immobilise their prey, in which case the chemicals are called ‘venom’. Snakes and spiders are probably the groups with the best-known examples of venomous predators. However, in marine invertebrates, venom also appears to be very popular, for example in the cone snails (*Conus sp.*), the greater blue-ringed octopus (*Hapalochlaena lunulata*, which applies the

⁵As early as the Middle Ages, blue cheese (with *Penicillium* fungi) was presumably used to fight infections. It is however not clear to me if the medicinal value was due to penicillin or due to some other component of the cheese.

same tetrodotoxin as the puffer fish for offence rather than defence), and almost the entire phylum *Cnidaria* (that contains jellyfish and sea anemones). The peripatus or velvet worm (*Euperipatoides kanangrensis*) applies a sticky secretion to catch its prey, a secretion which, surprisingly, contains the surfactant nonylphenol [32], an industrial pollutant of particular concern. The use of toxins in an offensive manner is not restricted to animals. The bacterium *Clostridium botulinum* produces extremely potent neurotoxins (often designated the ‘most toxic compound known’, for vertebrates), responsible for botulism. Apparently, the bacterium is using these toxins to kill larger organisms that are subsequently used as a source of nutrients. Interestingly, the same compounds are also used cosmetically as ‘botox’ injections to remove wrinkles by paralysing the facial muscles.



Figure 1.3: A collection of box jellyfish (class Cubozoa). Several species in this class produce a very potent venom, extremely painful, and sometimes even fatal for humans. This drawing from Ernst Haeckel’s “Kunstformen der Natur” (1904).

Sometimes, organisms release chemicals as a by-product of their normal metabolism, which can be toxic to other species. One example is the oxygen produced by photosynthesising organisms, as we have seen above. Another extreme example are the cave-dwelling bacteria that form ‘snottites’, which produce sulphuric acid at a pH close to zero. Even a naturally occurring physiological process like ageing is generally assumed to relate to the toxic action of reactive oxygen species, produced as a by-product of aerobic metabolism (this will be treated in Section 5.5). The chemicals (inadvertently) excreted by a predatory organism, can be used by prey organisms as a cue to alter their

morphology and/or life-history traits (reviewed by [173]). For example, ‘kairomones’ excreted by fish have been found to shift the life history of water fleas to smaller body sizes and earlier reproduction. This makes evolutionary sense as fish tend to prefer larger water fleas as food. One species (*Daphnia cucullata*) is known to grow a helmet and spine as a response to the presence of predators. A general discussion, and more example of changes in morphology and life history can be found in [98].

The role of humans

With the short, and perhaps somewhat random, overview above, I hope to clarify that chemical stress is a natural phenomenon. Organisms are continuously involved in an intricate chemical warfare with attack and counter-attack measures. Therefore, organisms have evolved systems to cope with such stresses to some extent. However, one particular species of mammal, *Homo sapiens*, is currently testing these defence systems (including their own) in ways that the earth has never seen before. Not only did we distribute naturally occurring chemicals such as heavy metals in more available forms around the globe, we also developed an enormous array of synthetic chemicals to support our way of living. We developed pharmaceuticals to cure ailments, pesticides to protect our crops, surfactants to clean our clothes, flame retardants to protect us from fire, and the list continues. Chemical industry is big, very big, business. Total production value of 2023 was estimated at 5.2 trillion Euro (that is 52 followed by 11 zeros).⁶

Despite the obvious benefits of synthetic chemicals, and their substantial contribution to our economy and lifestyle, there are side effects. Inevitably, most of these chemicals (or their breakdown products) will enter the environment where they interact with organisms. Organisms are able to deal with chemical stress, and can develop defence mechanisms, as the rise of resistance towards antibiotics and pesticides has shown. However, the problem with synthetic chemicals is one of scale; the sheer number of chemicals with very different structures and properties, the spatial scales that are affected (for a number of chemicals that means the global scale), and the time scales that are involved. Evolution requires time, and generally plenty of it. So the evolution of defence against synthetic chemicals is only an option for organisms with very short generation times and flexibility in their physiological basis. And, there is no such thing as a free lunch; adaptations come with costs for the organism, such as a reduced tolerance against other stresses, or decreased reproductive output. Furthermore, we should realise that we are also exposing our own bodies to these chemicals, not only directly but also indirectly as these chemicals interact with organisms that we feed on (crops and livestock).

This is not a book about the dangers of the chemicals we humans have been, and are still, using. However, I certainly do not consider this to be a trivial issue: human activities are placing an excessive burden on the planet, and chemical pollution is an important part of that burden. This is a book about understanding and predicting how chemicals impact individual organisms. My hope is that such an understanding will help to evolve the (still quite juvenile) science of ecotoxicology, and that it can eventually also play a part in reducing the chemical pressure on ecosystems (although

⁶Information from Cefic, The European Chemical Industry Council, at <https://cefic.org/a-pillar-of-the-european-economy/facts-and-figures-of-the-european-chemical-industry/>, accessed 18 June 2025.

I certainly hope that humanity will not wait for ‘more research’ before jumping into action). The point that I wanted to stress in the text above is that toxicity is an integral part of life. There is no scientific reason to distinguish between man-made synthetic chemicals and naturally-occurring compounds, poisons and venoms. All chemicals are potentially toxic, and some natural compounds have actually evolved for their toxicity.⁷ Furthermore, responses to toxic pressure in organisms share many features with effects of non-chemical stresses (e.g., radiation and temperature stress). Therefore, different, traditionally well-isolated, fields of research can learn from each other, and apply the same or similar modelling approaches as laid out in the remainder of this book. The first topic I need to address next is the matter of ‘complexity’.

1.3 Approaches to deal with complexity

Studying the effects of chemicals on biological systems requires dealing with a mind-boggling complexity. The number of chemical elements is limited (a bit more than a hundred), but they can be combined into a seemingly limitless number of chemical compounds. The last time I looked, the Chemical Abstract Service (CAS) registry held 219 million organic and inorganic substances, of which some 500,000 are commonly used and/or regulated across the globe.⁸ The number of species of organism is equally staggering: almost 1.7 million species of animal have been described,⁹ and that is just the animals, and only the identified ones. The total number of species on earth will be many times larger. To continue adding complexity, even the humblest of bacteria is a complex biological system. This system employs an enormous range of compounds such as DNA, RNA, proteins, lipids, carbohydrates, that are all intricately linked together to yield a working, adapting and procreating creature. How can we deal with the massive complexity of chemical stress? There are three common strategies in science to deal with complex systems that I discuss below, in an admittedly somewhat caricatural manner.¹⁰

The ‘black-box’ approach

In this strategy, one does not even attempt to understand the workings of the system. Instead, the researcher offers an input to the system (e.g., a chemical stress in a certain setting) and observes the output (e.g., the effects on a life-history trait). This is the traditional way of studying chemical stress on organisms, and the result is a description of the effect, after a specified exposure time, under the specific test conditions.

To illustrate this strategy with a practical example from ecotoxicology, consider the OECD standard protocol for a 21-day *Daphnia magna* (water flea) reproduction test [215]. The test starts with neonates (juveniles less than 24 hours old). After 21 days, the neonates have developed into reproducing adults, and the total offspring

⁷As an aside, I think that arguing to replace synthetic chemicals by natural ones is thus, in general, misguided, just like arguing for a ‘toxic-free environment’. Although the intentions are good, this latter concept needs a more practical definition to be helpful (see also [251]).

⁸Information from <https://www.cas.org/cas-data/cas-registry> and <https://commonchemistry.cas.org/> on 18 June 2025.

⁹Information from <https://currentresults.com/Environment-Facts/Plants-Animals/number-species.php> on 18 June 2025.

¹⁰Hendriks [93] presents similar thoughts, and uses art forms to explain different strategies.



Figure 1.4: What makes a duck, is it the exterior? Chromolithography from the series “Jagdbare Thiere” of Carl Friedrich Deiker (date ca. 1875).

production is counted. Next, the concentration estimated that is associated with, for example, 50% effect, relative to the control (EC50). If the answer is this EC50, what was the question that we started out with? Perhaps this is not so obvious, but we did not answer a question about *the* toxicity of the compound. Instead, we answered the question “What is the concentration that leads to 50% reduction of cumulative reproduction for *Daphnia magna* after 21-day constant exposure to chemical A, under standardised environmental conditions?” Assuming that the statistical analysis was adequate, we have the correct answer to a trivial question. What does this EC50 tell us about effects at any other timepoint than 21 days, or about another endpoint (such as growth or survival), or other test conditions (time-varying exposure, food limitation, etc.)? Very little to nothing. EC50s and similar statistics describe the effect after a specific exposure duration for the specific conditions of the test. Nothing more and nothing less.

In this crude form, the black-box approach cannot provide insight into the effects of a chemical. We have severely restricted the possible inputs by enforcing ‘standard’ test conditions and constant exposure, and we focus on a small subset of the outputs of the system (a single endpoint after a fixed test duration). However, the purpose of such a standard toxicity test is not to provide insights. This procedure is followed for rather administrative purposes: to provide a single number that can be used in environmental risk assessment, and that can be stored into a database. Rigorous standardisation tries to ensure that this number does not depend on the laboratory where it was determined. These databases with numbers are then used to investigate patterns of toxicity among chemicals and patterns of sensitivity among species.¹¹ This is not without dangers, given that the EC50 (and similar descriptive statistics) depends on (quite arbitrarily selected) exposure time, endpoint and environmental conditions in the experimental

¹¹These investigations rely heavily on statistical techniques such as (multivariate) regressions and principal component analysis (and, more recently, machine learning). Application of such techniques appears when we want to make sense of large amounts of data, but have no idea about the underlying mechanisms (and do not want to make assumptions about it). It is therefore not surprising that these techniques feature prominently in the ‘black-box’ strategy, in various fields.

test [127, 3, 102].¹² What makes matters worse is that these dependencies can differ radically between chemicals and species [3].

Less crude forms of the black-box strategy follow the chemical effects on more life-history traits over (a substantial part of) the life cycle. In this case, we follow more of the possible outputs of the box. If we also systematically vary the inputs, e.g., by varying an environmental factor, we can probe the black box in such a way that it allows an insight into its behaviour. This is a far more scientific attitude than the ‘standard testing’ (see [7] for discussion). However, coaxing useful information from the box in this manner is a painstakingly slow and resource-intensive process. For every species-chemical combination that we are interested in, we have to set up life-cycle experiments with a range of environmental factors. Furthermore, we also have to consider that chemical exposure in the real world is in the form of complex mixtures, and environmental factors do not change one at a time.

The black-box strategy is the most common one in ecotoxicology (at least for toxic effects), is taught in all ecotoxicological textbooks, and forms the basis of all of the regulatory frameworks for chemicals around the globe. The ‘standard testing’ is, at least from a scientific perspective, largely a waste of time, resources and organisms. Life-cycle testing is useful, but the road to understanding and generalisation is extremely slow. In my opinion, chemical-stress science needs to progress beyond the black box and focus attention on the box’s innards.

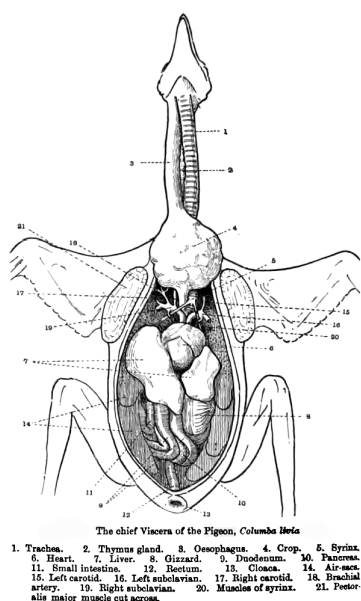


Figure 1.5: What makes a duck, is it the interior? Well, this picture represents a pigeon, actually. From the textbook “Zoology; an elementary text-book” of Sir Arthur Everett Shipley (1901).

¹²The value of the EC50 even depends on how the endpoint is determined. Choosing body length or body weight yields a different value, and so do reproduction rate and cumulative reproduction. In 2011 [102], I summarised the most important problems of the EC_x approach, and made a plea to move away from this kind of summary statistics. Alas to no avail, so in 2025, I again clarified why such a move is needed [112].

The ‘white-box’ approach

In software engineering, a white box is a system to be tested, where all internal components can be viewed (but usually not altered). The ecological analog is an approach where one attempts to determine *all* of the elements that are important for the behaviour of (a part of) the biological system. This is scientific reductionism at its best: the view that a system can be understood when we identify all of its components and how they are connected. In a chemical-effects setting, this strategy was probably initiated in pharmacology, where researchers looked for the receptors that interact with chemical compounds such as drugs, and the links between these receptors and the rest of the body. In ecotoxicology, there also is an important line of research that tries to identify the enzymes responsible for transformation of compounds, and the receptors that link to the observable effects.

As an illustration, consider the mechanism of action of organophosphate pesticides. To understand how these compounds affect organisms, researchers have identified a whole chain of events. First, the chemical is taken up into the body, where it is transformed into the ‘oxon’ form by an enzyme. The oxon, if it survives other metabolic conversions, binds to the enzyme acetylcholinesterase, making it dysfunctional. Without this enzyme, the neurotransmitter acetylcholine is no longer removed from the synaptic cleft, which means that the nerve at the other end of the cleft remains stimulated, even when the original stimulus is long gone. This overstimulation causes effects, and eventually kills the animal. Resolving this mechanism of action has greatly supported the design of new pesticides.

A more recent incarnation of this strategy involves the so called ‘-omics’ or ‘systems biology’ approaches; focussing on the molecular and cellular level (see e.g., [256, 6]). In ecotoxicology, these approaches generally focus on mapping the molecular-level changes that result from exposure to a certain chemical. These can be changes in gene expression (genomics), RNA transcription (transcriptomics) or the proteins (proteomics) and metabolites formed (metabolomics). Obviously, such methods yield an enormous amount of output. To make sense of these data requires statistical techniques.¹³

The white-box strategy has been immensely useful to understand the mechanistic aspects of chemical effects on organisms. The success of this approach is probably best illustrated by its commercial applications: many pharmaceuticals and pesticides have been purposefully designed to perform a certain action in a biological system. However, there are limitations to consider, in view of the scope that I laid out in Section 1.1. For example, the link between the molecular level and life-history traits such as growth and reproduction is still an extremely vague one [6, 84]. This is not surprising as growth and reproduction are the result of a host of interacting metabolic processes, and they are subject to numerous feedbacks. For example, body size determines the feeding rate, and thereby also the resources available for reproduction. A chemical that affects the growth process will therefore, inevitably, affect reproduction as well. Such feedback loops do not emerge from the molecular level but rather from energetic constraints on the whole organism. It would be naive to assume that effects on life-history traits can always be reconstructed from a direct series of changes at the molecular level only [106].

¹³This shows that (at least in stress ecology) the -omics approaches are, at this moment, actually not so different from the ‘black-box’ strategy after all. If we would know what all these molecules were doing, we would not need these statistical techniques. The term ‘bioinformatics’ is nowadays almost exclusively used to denote this type of analysis.

Even if we, one day, can fully link the changes at the molecular level to effects on the life history, the result will still be highly species and chemical specific. In other words, generalisation that I aimed for in Section 1.1 would remain illusive. When a white box is extremely complex, it might as well be a black box. For human toxicology, however, a white-box model might be a feasible and useful project. In this field, one is only interested in a single species, for which there is a lot of research interest and funding. Furthermore, the interest is not in highly integrated endpoints such as growth and reproduction but in much more subtle, low-level, effects such as hormone changes and tumour induction. A ‘virtual human’ [262] might therefore be within reach. As Yang and co-workers write [262]:

In pursuing an understanding of toxicological risk, it is impractical to investigate all of the vast number of combinations of chemical mixtures and multiple stressors because the numbers approach infinity. A more realistic strategy is to focus on finite number of entities; thus, we should investigate biological processes which, though complex, represent finite entities.

However, the type of investments in human physiology, molecular biology, and toxicology are unthinkable for the vast majority of creepy crawlies that biologists and ecotoxicologists deal with. We have to consider more efficient approaches, and in my opinion, we can learn in this respect from other (more mature) scientific fields.

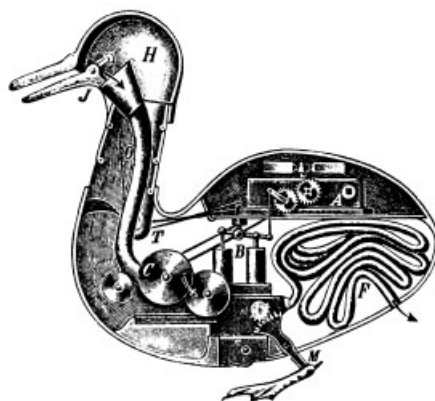


Figure 1.6: What makes a duck, is it the mechanics? A postulated interior of the Digesting Duck of Vaucanson (1738-1739) by an American observer.

The ‘simple-box’ approach

In the ‘simple-box’ strategy,¹⁴ we are not going to deal with the system itself, but make an idealisation that we can study. In biological specialisations, such a strategy is less common than in fields such as chemistry, physics and engineering. As an extreme example, physicists can explain and predict the orbits of celestial objects with great accuracy, without worrying about the compounds that make up such a body. They do this by making an idealisation; the only thing that matters is an object’s mass, which

¹⁴I borrowed this term from my former colleague Dik van de Meent who has developed a chemical fate model by the name of SimpleBox, which indeed follows this strategy. See RIVM report available from <https://rivm.openrepository.com/entities/publication/5c00b8d8-72d0-4337-8398-03f3d5009ddb>.

can be treated as focussed in a single point of infinitesimal size. Similarly, environmental chemists simplify the world when they explain and predict the distribution of chemicals emitted into the environment. In these environmental fate models, heroic simplifications are made, for example assuming that large parts of the environment can be treated as well-mixed homogeneous boxes (e.g., [183]). There is no need to characterise all of the chemicals that make up the environment to make a useful prediction of environmental fate. Such idealised models are thus (relatively) simple and efficient, and extensively used in chemical risk assessment [185].

Interestingly, there *is* a branch of ecotoxicology that heavily relies on this strategy: toxicokinetics. Toxicokinetics deals with the processes of uptake, distribution, transformation and elimination of chemicals in organisms. In other words, it provides a causal link between external levels of a chemical and internal ones (thus, in a sense, a fate model for an organism). In this field, the common approach is to simplify the organism to one or a few well-mixed boxes. These researchers are well aware that organisms are complex systems, with an enormous range of components that affect the chemical's fate, but they choose to bring this system back to its bare essentials. Toxicokinetic models are commonly applied in ecotoxicology to analyse and predict internal concentrations, but also in pharmacology to predict the delivery of drugs to specific parts of the body (in that case often referred to as PBPK models).

A limitation of this strategy is that, working with a simplification, we are always wrong. A model is an idealised representation of reality, which implies that it cannot capture everything. Is that bad? That depends on the question. A physicist may simplify the earth to a point mass, which is unacceptable for a geologist. A more serious problem is that you would have to know quite a bit about a system to simplify it successfully, or to trust someone else's simplification. It is all too easy to come up with complete nonsense from a plausible-sounding simplification. An example from history may illustrate the problem. William Thomson (Lord Kelvin) in the 1860's estimated the age of the earth at 20-400 million years, based on the simplification that the earth started as a completely molten object, which has subsequently cooled down to its current temperature. This estimate did not include heat convection inside the earth, nor the release of heat by radioactive decay (an unknown process in the 19th century). Scientists now set the age of the earth at 4.5 billion years. The crucial point in the development of an idealised model lies in the formulation of simplifying assumptions. At the very least, these assumptions must be clearly and explicitly presented, and the set of assumptions should be internally consistent and consistent with established scientific theory. The actual model should follow from these assumptions in a unique manner. Much of the wariness surrounding biological modelling, and the use of model results in ecotoxicology and risk assessment, may come from assumptions being hidden or poorly communicated.¹⁵ However, it should be noted that some form of 'model' is also used in the black- and white-box approaches, but in those cases they are extremely silly (as descriptive statistical models in the black-box approach) or extremely complicated with hundreds of (implied) assumptions (in the white-box approaches).

A key question in the 'simple-box' strategy is: how much should we simplify the system? Simplify too little and we run into the same problems as in the 'white-box'

¹⁵The environmental chemists apparently did a good job in embodying trust in their fate models as these models are a mainstay of chemical risk assessment. However, my guess is that this is more based on 'authority' than on a clear communication of assumptions.

strategy. Simplify too much and we may end up with a ‘black box’ again. In the end, the degree of simplification depends on the question that you want to address; “as simple as possible, but as complicated as needed.” Or, as Albert Einstein put it:

It can scarcely be denied that the supreme goal of all theory is to make the irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience.

Biologists are not used to make idealisations of the functioning of individual organisms, although they do for populations.¹⁶ Biology is traditionally a study of details, complexity and exceptions. However, the simplification strategy has worked in other fields to deal with complexity, and is therefore the best candidate strategy for chemical-stress ecology, given the scope I laid down in Section 1.1. It is good to realise, however, that simplification is usually well served by the information gathered using the black- or white-box strategies.

1.4 Can we simplify biological systems?

It seems obvious that we do not need to know all of the molecules in an organism to understand and predict toxic effects. But to what extent can we simplify biological systems without losing essential behaviour in their response to stressors? How simple can a general quantitative theory be and still capture the essential elements of an organism’s life history, and the effects on toxicants on the life-history traits? From an idealised system, can we understand why toxic effects change in time, vary between species and toxicants, and depend on environmental conditions? Can we understand the effects of mixtures of toxicants or the interactions between toxicants and other stressors such as food limitation? And, can we ultimately use these idealisations to produce reliable predictions for environmental risk assessment? I certainly do not claim that all these questions have a satisfactory answer at this point. However, in the remainder of this book I will outline a theoretical framework for chemical stress ecology in which such questions can be studied in a formal manner, and I will summarise the achievements that have been made up till now.

As explained earlier, I will focus on the effects of chemicals on the individual’s life-history traits, such as growth, development, reproduction, and mortality. These traits are directly relevant for the population level, and are easily observable in experimental tests. Furthermore, the level of the individual is of special interest because we can make natural use of the conservation laws for mass and energy (one of the few ‘hard’ laws that can be applied in biology). At the population and molecular level, the application of these laws is much harder because the boundaries of the system are much more vague. As we will see in later chapters, the focus on individuals will help to interpret effects at lower levels of organisation, and helps to build models at the population level (and higher).

The framework that we are looking for should explain the links between these traits, and explain the effects of toxicants and other stressors over the entire life cycle (from

¹⁶Population modelling has a long tradition in biology, and indeed relies heavily on simplification. Take the example of the Lotka-Volterra system: https://en.wikipedia.org/wiki/Lotka%E2%80%93Volterra_equations. For many biology students, this classic model will be their introduction into modelling.

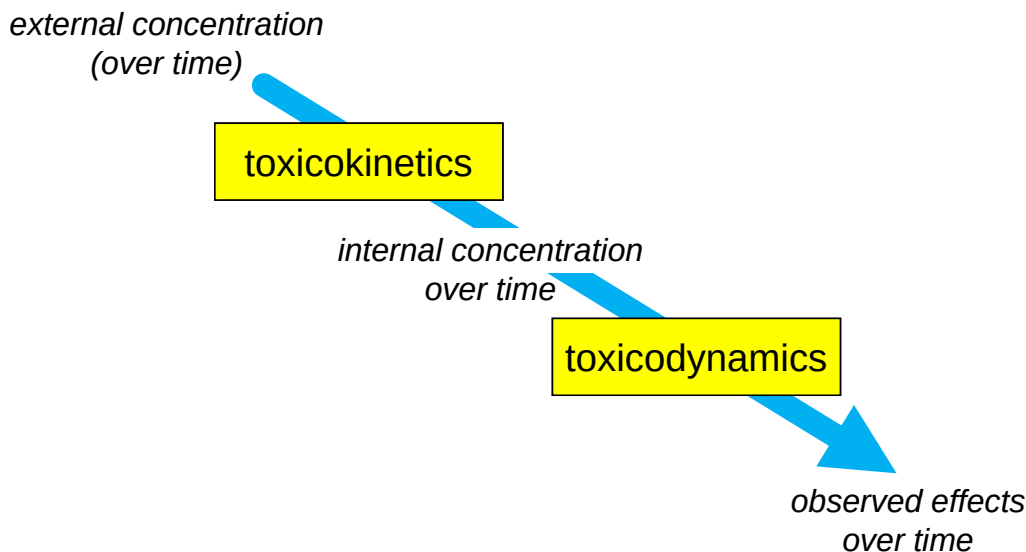


Figure 1.7: The basic approach towards process-based modelling for effects of toxicants. Toxicokinetic and toxicodynamic models are used to translate (time-varying) external concentrations to time patterns of effects.

egg to death). It should allow prediction of effects under untested conditions, such as time-varying exposure scenarios or different environmental conditions. Therefore, the approach should be ‘mechanistic’, i.e., all parameters should have a physical interpretation. And finally, it should be as simple as possible to be useful. We need an approach that has as few species- and compound-specific assumptions as possible, to avoid having to develop a new framework for each and every combination. From these requirements, it follows that our framework can be classified as a toxicokinetic-toxicodynamic (TKTD) approach (see [10]).

1.5 Toxicokinetics and toxicodynamics

Simplifying chemical stress requires addressing both toxicokinetics and toxicodynamics (Fig. 1.7). It is generally (and possibly by definition) true that chemicals first need to be taken up into the body before they can exert a toxic effect. Modelling uptake, distribution and transformation of chemicals inside organisms is the field of toxicokinetics (TK). Thus, the first type of models we need to consider are TK models. A TK model will yield a prediction of the concentration of a toxicant (or a relevant metabolite) inside an organism, perhaps even in a particular organ. TK has been extensively studied in both ecotoxicology and pharmacology, and a range of modelling approaches exist (e.g., [23, 184, 238]). The simplest of these is the one-compartment model with first-order kinetics. The entire organism, with all its complexity, is simplified to one well-mixed compartment. Despite this heroic simplification, this model still manages to capture the general patterns to such an extent as to be useful. At the other end of the spectrum are physiologically-based pharmacokinetic (PBPK) models. These models include more biological detail by defining separate compartments for each organ, or group of organs, and specifying the blood flow through the body. Of course, such models are much more species specific, and are mainly developed for mammals, which is why I will not deal

with them in this book (see [229] for an overview, and [212] for a fish example).

The chemical, after being taken up, will interact with one or more ‘target sites’ to affect one or more processes in the organism. I am using the word ‘target site’ in its regular, somewhat abstract, meaning to indicate the part of the regular biology of the organism with which the chemical (or a metabolite) interacts to ultimately produce an effect on the life-history traits. This part of the organism can be a cell membrane (as assumed for the toxicity of narcotic or baseline chemicals), a host of macromolecules (as assumed for reactive compounds), specific enzymes (as assumed for organophosphate insecticides), etc. The models needed to translate the internal concentration at a target site to an effect on a trait over time are called toxicodynamic (TD) models. In contrast to TK, TD is studied much less in a mechanistic sense, and fewer modelling approaches exist. TK is not inherently simpler than TD, so I expect that this is a matter of tradition. TK modellers have adopted the scientific attitude common in chemistry and physics, and are therefore used to simplifying reality. In contrast, TD lies deeper in the realm of biology, and biologists seem more hesitant to simplify biological systems to such an extent.

How can we simplify biology to suit a TD model? Part of the simplification is to lump the biochemical processes affected by the toxicant under the header ‘damage’ (explained in detail in Chapter 3). However, we also need to simplify the biology of the organism at the individual level. As stated earlier, I look for simplifications that explain life-history traits such as growth and reproduction from underlying processes. But how deep do we need to go? We could try to build a model that contains organs (like in PBPK models), or go to the cell or even the molecular level. However, we would end up with an extremely complex model, with an enormous amount of parameters, that is also highly species specific. What are the fundamental processes common to all living organisms? All organisms need to take up resources (energy and building blocks) from their environment and use it for their processes of growth, development, maintenance and procreation. This is just as true for algal cells as for the largest whales. Furthermore, we know that all organisms abide the laws for mass and energy conservation. It therefore makes sense to focus on this level of complexity, the level of general metabolic organisation. This is the subject of Dynamic Energy Budget (DEB) theory. In the next chapter, I will discuss this theory in much more detail. In the next section, I will present some very general principles first.¹⁷

1.6 Principles of mechanistic modelling

The main purpose of modelling, as I see it, is to gain an understanding of the dominant processes that drive a system’s behaviour. Subsequently, this understanding can be used for predicting the system’s behaviour under untested conditions (extrapolation). The purpose of modelling should *not* be to draw a line through data points; that works much better without a model (just take a pen and connect the dots). Any model for chemical effects on life-history traits that claims to be ‘process-based’ or ‘mechanistic’ must have certain properties. Such a model must be able to explain effects on life-history traits in an integrated manner, over the life cycle (in principle from egg to old age). It must

¹⁷For more introduction into the basics and principles of mechanistic modelling, I suggest my own e-book on the topic [110]. The extensive review of [157] and the gentle introduction in [105] may also be useful in this context.

be able to deal with time-varying conditions (e.g., food or toxicant levels), and allow extrapolation to effects under untested environmental conditions (limiting food levels, other temperatures, etc.). It must thereby respect the conservation laws for mass and energy.

The distinction between ‘mechanistic’ and ‘descriptive’ is not always very strict. Mechanistic models usually contain descriptive elements, and descriptive models can apply mechanistic descriptors. As an example of the latter, a log-log regression of bioconcentration factors¹⁸ against the octanol-water partition coefficient¹⁹ is certainly a descriptive model. However, this descriptor was selected on the basis of assumptions about the underlying mechanism (a similarity between octanol and animal lipids). All mechanistic models follow from theory. A theory can be thought of as a consistent and coherent set of simplifying assumptions about the real world. Theory is specifically useful when it can be applied, which often requires a translation into a mathematical model. This allows the model to be calibrated and tested on experimental data. There are several general modelling principles that apply to any kind of model (well, at least the useful ones), which I will briefly discuss below.

- The models we are interested in here take the form of a dynamic system. The current state of the system is fully defined by a set of ‘state variables’. Once we know the current state of the system, and how it will change over time depending on the value of the state variables, we can predict its future development. The part “how it will change over time depending on the value of the state variables” indicates that we generally deal with (sets of) differential equations.²⁰ These differential equations have model parameters. In a simple TK model, the internal concentration of a toxicant can be a state variable. How it changes in time depends on the external concentration and on the value of the internal concentration, and requires model parameters such as rate constants.
- Model complexity should link to the question that needs to be addressed. Some questions require complex models, some require simple models. The fact that a system is complicated does not mean that we cannot describe essential behaviour with a simple model. Models must be as simple as possible, but as complex as they need to be. However, complex models (even mechanistic ones) can never lead to general insights.
- Model complexity is generally linked to the number of model parameters. The more parameters need to be estimated, the higher the demands on the data in terms of quality and/or quantity. When the data basis is insufficient to fit all model parameters, the model might be too complex, although it is often possible to use prior information of some sort (estimation routines, rules of thumb, parameters for related species or toxicants, etc.).

¹⁸The bioconcentration factor (BCF) is the ratio of the concentration in an organism to the concentration in water, when the concentrations do not change anymore (steady state). This metric plays a role in environmental risk assessment for chemicals.

¹⁹The partition coefficient between octanol and water (K_{ow}) is a proxy for the tendency of a compound to bioaccumulate from water into lipid phases of organisms. When given a choice, chemicals with a high K_{ow} tend to leave a water phase in preference of a non-polar phase such as lipids. This tendency is called hydrophobicity; literally a ‘fear of water’.

²⁰For a gentle introduction in differential equations, you can consult my e-book on modelling principles [110].

- The more specific a model is for a particular situation (e.g., for a particular species and toxicant), the less useful it is for other conditions. There is an enormous range of chemicals and species, and we cannot seriously consider making a new model for each combination. Apart from the investment of time and resources, it would also be impossible to compare species or chemicals, to extract general patterns, or to draw general conclusions.
- Models follow from assumptions. The translation from assumption to model should be one-on-one; that is, it should not involve any new (hidden) assumptions. If the model fails to predict the real world, one (or more) of the assumptions were not applicable (for this particular data set). If all the assumptions are made explicit, this means that we do not have to dive into the equations to judge the usefulness of a model. This sounds trivial, but it is not: explicitly presenting the underlying assumptions is still an exception in the documentation of models.
- Models are a simplification of a part of the real world, and therefore, models are always ‘wrong’. However, experimental data are not always ‘right’; they are subject to uncertainty as well, and require interpretation to arrive at a conclusion. Many things can and will go wrong in experiments. Therefore, when a promising model does not fit, it usually pays to scrutinise the experimental set up and execution first.
- For a model to make sense, its dimensions must be correct.²¹ A model with dimension problems cannot be mechanistic, is useless to increase our understanding, and generally does not allow useful extrapolation to untested situations. Dimensional correctness is necessary for a mechanistic model, but not sufficient.

²¹The physical dimensions that we usually encounter are mass, length, time, energy and temperature. It is not allowed to add or subtract entities that have a different dimension, and at both sides of an equal sign the same dimension must result. Units follow from dimensions and specify them further. The unit ‘gram’ is a particular choice for the dimension ‘mass’. Dimensional analysis is particularly useful as a quick test of a model.

Chapter 2

A model for energy budgets

2.1 Why do we need an energy budget?

Before we can consider toxic effects, we first have to look closer at the general biology of the organisms, and particularly its life cycle: how do organisms feed, grow, develop, reproduce and die? Toxicant effects are deviations from the situation without toxicants, and therefore, we first need a quantitative model for the unexposed reference situation before we can interpret toxic effects. But, how can we make sense of the staggering biological complexity of living organisms? Surely, we do not want to build a new model for each and every species, so the simplification must be a general one. Let's start by stripping away most of the complexity, and start from what is common to all organisms. All living organisms take up resources (energy and building blocks) from the environment, and use these resources to fuel their life cycle (Fig. 2.1). They need to maintain their integrity, grow, develop, and reproduce themselves, and in performing those activities, they need to obey the conservation laws for mass and energy. What we need is a set of general rules for the acquisition of these resources, and their division over the various sinks (the energy-requiring processes in the individual). This set of rules must be internally consistent, and consistent with general empirical observations. Further, the rules must always obey the conservation laws, and need to cover the complete life history (from embryo to death).

This set of rules cannot be exactly the same in all species. Chicken obtain their

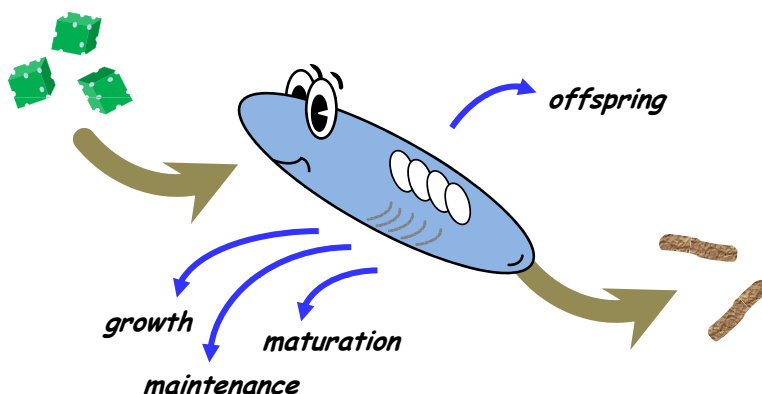


Figure 2.1: All organisms take up resources from the environment and use them to fuel their life histories.

energy and building blocks from a single flow of resources (food), and reproduce by producing eggs (each egg a tiny embryo surrounded by a big blob of reserve material). In contrast, a green alga like *Chlorella* can use energy from sunlight and building blocks from the nutrients in its environment, and reproduces through division. Although these organisms clearly cannot follow the exact same set of rules for their metabolic organisation, we can expect a logical relationship. All organisms are related through evolution, and thus, we must be able to come up with an evolutionary narrative to explain how different sets of rules have developed across all living organisms. These rules are so essential to the functioning of life that it makes sense to assume that they are well conserved across the lineages. It is likely that all higher plants follow the same set of rules, which should differ in a predictable manner from those of the green algae that they are related to. It is likely that all birds have the same set of general rules, and that this set differs only slightly from the set for reptiles, in a way that we can explain by evolution (e.g., birds spend a large amount of the ingested energy to heat their bodies, something that reptiles do not do).

What we need for understanding toxic effects on traits (see Fig. 2.1) is first and foremost an understanding of the rules for metabolic organisation, i.e., a model for the energy budget. We first need to understand how growth and reproduction are fuelled from food, before we can understand the action of stressors on these traits. Fortunately, we do not have to develop something from scratch. A formal theoretical framework for metabolic organisation in all living organisms already exists: Dynamic Energy Budget (DEB) theory. This theory is the brainchild of Bas Kooijman, who started the development in 1979. Since then, the theory has been formalised [240, 241], and its applications have radiated in such diverse directions as ecophysiology, (eco)toxicology, evolution, aquaculture and isotope dynamics. For more background of the theory, I would like to refer the interested reader to the book by Bas Kooijman [155] or the conceptual papers in the open literature [154, 213, 143].¹

In this book, I will focus on animals, in particular ectotherms, reproducing through eggs (including ovo-viviparous animals²). These animals feature prominently in ecotoxicological research and form, metabolically speaking, a rather homogeneous group. All of them feed on (materials from) other organisms to obtain their energy and building blocks, which places similar constraints on metabolic organisation. This similarity is for example supported by the observation that growth curves for a wide range of animals (and some other organisms too) is well described by the same curve (under constant conditions): the von Bertalanffy growth curve [155]. Focussing on ectothermic animals only, allows me to present the principles and consequences of DEB theory in the clearest manner possible; it is my experience that the full width of DEB theory can be overwhelming.³ However, it is good to realise that DEB theory aspires to cover *all* living organisms on this planet (and perhaps even on others). The ectothermic animals form just a small part of all of the metabolic types that have been realised by evolution. However, a thorough understanding of these animals forms a solid basis for further exploration of the theory. Our starting point is formed by the ‘standard DEB

¹For a list of DEB papers up to 2017, see http://www.bio.vu.nl/thb/deb/DEB_papers.pdf. More recently, DEB papers are included into a Zotero library, see the DEB portal: <https://debportal.debtheory.org/docs/>.

²This means that the eggs are retained within the mother until hatching. The nourishment for the embryo comes from the egg only; there is no placental connection. Well known examples are the water flea (*Daphnia magna*), guppy (*Poecilia reticulata*), and woodlouse (*Porcellio scaber*).

³Endotherms have some specific issues in DEB theory, which will be shortly discussed in Section 2.7.

animal model’ [167, 240, 241], which concerns an animal that does not change shape over growth, feeds on a single type of food with a constant composition, and reproduces by means of eggs.

2.2 Concepts in the theory

In this section, I will explain the major concepts in DEB theory. This section is intended to help you understand how these concepts work. Because assumptions play a central role in this line of work, I will highlight the required assumptions by placing them in boxes. A thorough and critical discussion of why these assumptions are used in the theory (and which alternatives are possible) is largely outside the scope of this book, although I will briefly provide the logic behind each of them. It is good to realise that DEB theory is not beyond criticism; in fact, there are several aspects that require further detailed investigation, in my opinion. Nevertheless, it is a general and consistent framework for the energy budget, and currently the most well-developed and best-tested framework available for our purpose.

Energy as central currency

Because we are interested in processes at the life-history level, it makes no sense to focus on individual chemical compounds in the food and in the body. Such a level of detail would lead to highly complex, and highly species-specific models. For that reason, DEB theory focusses on energy as a common currency (with units of e.g., Joule). Mass and energy flows in an animal are linked. A certain mass of food represents a certain amount of energy, depending on its composition. The growth process represents an amount of energy obtained from food that is fixed in the new tissue, and energy that is lost in the process of turning food into new structure (overhead costs, which need to be accounted for in any biochemical transformation). The theory allows following individual chemical elements, but this is substantially more complex, and not essential for the applications that I discuss here.

Volumes and surfaces

The interplay between volumes and surfaces plays a central role in DEB theory, so it is good to focus a bit on geometry. Figure 2.2 shows three basic shapes and shows how their volume and surface area depend on one or two particular size measures (height and/or radius, which are length measures). Imagine that we increase an object in size but keep its proportions the same. The cube remains a cube, the sphere a sphere, and the cylinder a cylinder with the same factor between radius and height. Such changes in size are termed ‘isomorphic’ (literally ‘same shape’). In the cube, if we increase height by a factor of 2 in an isomorphic change, the area increases by a factor of 4 and the volume by a factor of 8. The same happens with the sphere for a change in the radius. What would happen if we focus on another length measure such as diameter or circumference? We can rewrite the equations in Figure 2.2 for this new length measure (diameter is two times the radius, circumference is two times π times radius). You can do the math yourself, if you like, but you will find that volume will always be proportional to the length measure cubed, and surface area to the length

measure squared. The proportionality constant will depend on the length measure that we selected. For the cylinder to change isomorphically, there has to be a fixed factor between radius and height ($h/r = \text{constant}$). We can thus rewrite the equations for volume and surface in Figure 2.2 to contain only length *or* radius. Just as with the sphere, volume will always be proportional to the length measure cubed, and surface area to the length measure squared.

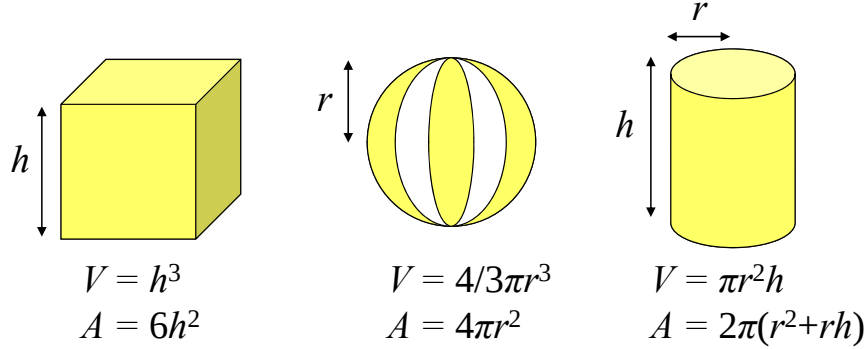


Figure 2.2: Different shapes with their volumes V and surface areas A ; h is height, r is radius.

The morale of this story? For any three-dimensional shape that changes size isomorphically, its volume scales with cubed length and its surface with a squared length. This is true independent of how we take the length measure. However, if we want to calculate the actual area or volume from a length measure, we need to consider the shape. The cubic root of the volume gives a ‘volumetric length’. For the cube, the cubic root of the volume gives exactly the length of the edge. So, for any shape, the cubic root of volume provides the edge of a cube with the same volume as the original object. In a DEB context, we will often use ‘structural length’ as a state variable, which is the volumetric length of the structural component of biomass (explained in a later section). The ratio between the volumetric structural length and an actual measure of length is called the ‘shape-correction coefficient’. This coefficient depends on the shape of the object, but also on the way the length measure is taken. For the cube, the coefficient is 1 if we consider the length of an edge; for the others it can be calculated using the cube root of the volume. For the sphere, for example, the shape correction coefficient is $(\frac{4}{3}\pi)^{1/3} \approx 1.61$ when length is quantified as the radius, and $(\frac{\pi}{6})^{1/3} \approx 0.81$ when length is expressed as diameter. The same principles do not only hold for nice geometric objects, but also for organisms, as illustrated in Figure 2.3.

The interplay between volumes and surfaces is particularly important in DEB theory because uptake of food and toxicants is across a surface area whereas maintenance processes and concentrations of toxicants are associated with a volume of biomass. In the standard DEB model, we assume that animals grow isomorphically, so they do not change their shape when they grow. In detail, this assumption almost never holds, but it works very well as an approximation for a broad range of animals. It should be noted that isomorphy is an assumption of the standard model, not of DEB theory itself; the theory deals with surface areas and volumes, which requires just a bit more thought for non-isomorphic growth.

Assumption: the standard animal grows isomorphically; i.e., it does not change shape over its life cycle.

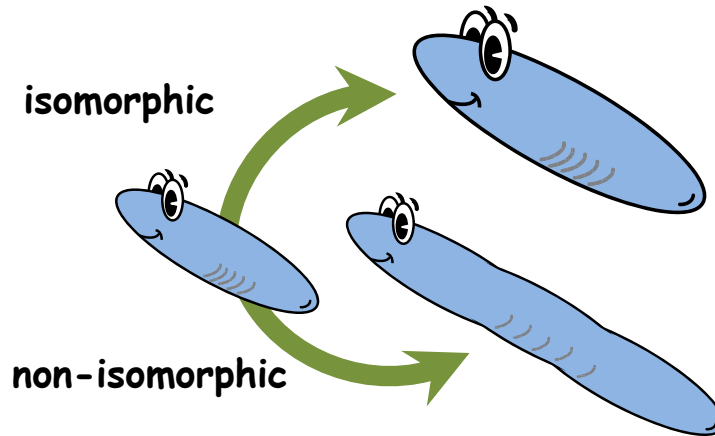
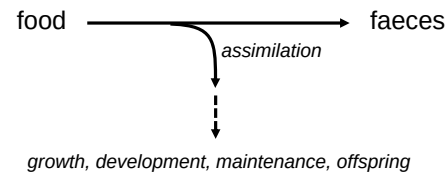


Figure 2.3: Isomorphic and non-isomorphic changes in shape with growth.

Feeding

Feeding involves the transport of food particles from the environment into the organism across a surface area. Therefore, we take the ingestion rate proportional to the structural surface area (structure is defined in the next section). The proportionality constant is species- and diet-specific. This simple assumption generally fits empirical observations of size-related feeding rates very well (see [155, 265]).



Assumption: the feeding rate is proportional to the structural surface area of the organism.

Ingestion rates depend on the individual's body size, but also on the food density in the environment. This relationship is called the 'functional response' in the ecological literature. In the standard model, we assume that the total time budget is used to search for food, or to handle it. This leads to an ingestion rate that follows a hyperbolic relation in food density, which is also known as the Holling type II functional response.

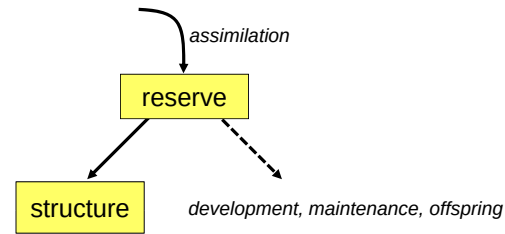
Assumption: the organism is either searching for food or handling it, and handling time does not depend on food density.

The standard model does not deal with the details of the digestion process; the energy from food is immediately assimilated for further use with a certain transformation efficiency (again, overhead costs need to be paid). The processes in the gut can generally be ignored as they are fast relative to the life-history traits that we are primarily interested in. However, when we want to follow the uptake of toxicants through food, we may want to take a closer look at what happens in the gut.

Assumption: a fixed fraction of the energy from food is instantaneously assimilated for further use. This implies a constant assimilation efficiency.

Structure and reserve

Biomass is made up of an enormous amount of different chemical components. Clearly, following all chemicals inside an organism does not lead to a workable model. Therefore, we lump all these components into two categories: structure and reserve. Structure requires maintenance, reserve does not; reserve fuels metabolic processes, structure does not (only in case of rather severe starvation, which is a deviation from the standard rules). Why do we need reserve? The presence of a reserve allows the organism's metabolism to be buffered from a non-constant arrival of food particles. That way, the individual is not immediately in a starvation situation when it does not feed continuously (although the gut may also provide some buffering capacity as well). Furthermore, delineation of two biomass components can be used to explain changes in body composition.



The presence of reserve can also explain why eggs initially do not use oxygen, but during development *decrease* in total mass while they *increase* in respiration. The freshly-laid egg consists almost completely of reserve, which does not require oxygen. During development, structure increases and the reserve decreases, which leads to a rise in respiration per mass of egg. In DEB theory, the storage of resources in the egg (e.g., yolk) is treated as an integral part of the reserve (sharing the composition and dynamics of the internalised reserve). I consider this to be a tricky assumption; the fact that eggs have a (non-respiring) storage does not provide a strong argument for the presence of a reserve compartment in juveniles and adults.

It must be stressed that these two categories are abstractions; you cannot take an organism and measure its reserve and structure directly, or point at a specific chemical and put it in either group. One type of compound can be classified as a reserve component, a structural component, or a mixture of both. The only way to decide to what extent a specific compound should be classified as reserve or structure is from its dynamics: e.g., how its density changes under different food densities or under starvation. It must also be stressed that ‘reserve’ in the DEB context does not imply that these compounds are inert and set aside for later use; reserve components can have active metabolic functions.⁴ A compound class like ‘lipids’ belongs to both reserve and structure, and this division depends on the species (not all animals rely on lipids for storage as heavily as humans do).

Assumption: there are two types of biomass; reserve which fuels metabolism, and structure which requires maintenance. The standard animal has one reserve and one structure compartment.

In the standard DEB animal model, there is one structure and one reserve. If necessary, this can however be extended to more structures and more reserves.⁵ Furthermore,

⁴As an example, RNA should be classified (at least largely) as a reserve component because of its dynamics: RNA content is high at high growth rates, which occur when reserve density is high. Following the same reasoning, DNA should be classified as structure: the amount of DNA per unit of cell weight decreases with increasing food densities as the weight of the cell increases due to the build up of reserve.

⁵Even models without reserve can be useful, as discussed in more detail elsewhere [132, 109, 107] and see Section 2.6.

there are two more categories that can add to the total bulk of an organism: a reproduction buffer (dealt with later in this chapter), and products, which do not require maintenance but also cannot be used to fuel metabolic processes (e.g., shells, hair, exoskeletons). The extension with products is outside the scope of this book, but is dealt with in [155, 219].

Homeostasis

Homeostasis is the property of a system to maintain a constant condition. For an organism, the ability to maintain a constant internal environment is clearly advantageous because it makes it easier to run a metabolism. DEB theory makes use of homeostasis in several of its assumption. One of the most important ones is ‘strong homeostasis’: structure and reserve have a constant chemical composition. This means that a unit of reserve has a constant ability to do work, and that a unit of structure always costs the same amount of energy to produce. The composition of reserve will differ from that of structure, which means that when the reserve-to-structure ratio varies as a result of changes in food availability, the total composition of the body will change. As long as the strong homeostasis assumption holds, we can thus use the variation in body composition to tell us something about the ratio of reserve to structure.

The strong homeostasis assumption is a key property of the theory because it allows us to focus on two abstract entities (reserve and structure) instead of the whole range of individual chemicals that make up biomass. If we want to deal with changes in composition that are not matched by the standard model, it is better to increase the number of structures and/or reserves than to modify strong homeostasis.

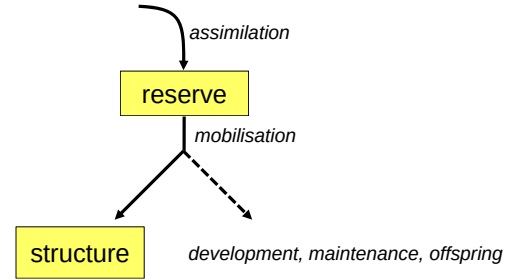
Assumption: reserve and structure have a strictly constant composition (strong homeostasis).

Additionally, DEB theory applies the assumption of ‘weak homeostasis’: under constant food density, the composition of the total body becomes constant, and will remain constant while the animal grows. Together, weak and strong homeostasis imply that under constant food densities, the ratio between the amount of reserve and the amount of structure (reserve density) becomes constant. Weak homeostasis relies on strong homeostasis, and is weaker (hence the name): it only applies to constant environmental conditions. However, this assumption is important as it fixes the rate at which reserve is made available for further use. Weak homeostasis limits our freedom to come up with alternative specifications for reserve dynamics. The weak and strong homeostasis assumptions, together with the distinction between structure and reserve, constitute a huge simplification of biology, while still allowing for a range of realistic behaviours.

Assumption: at constant food density, the reserve will (after an equilibration time) be a constant proportion of structure from birth to death (weak homeostasis).

Reserve mobilisation

Energy needs to be mobilised from the reserve to fuel all activities of the organism. The rate at which reserve is mobilised will likely vary over time, but how can we specify this rate? As a first constraint, assume that mobilisation depends only on properties of the individual (such as structural body size and available reserve), but not on external factors such as food availability. After that, we have the assumption of weak homeostasis to constrain the options for the reserve dynamics considerably.⁶



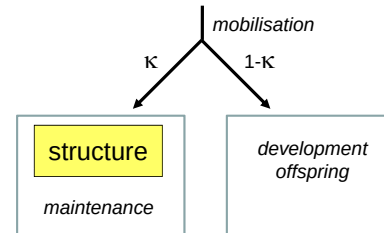
Assumption: The mobilisation rate of the reserve depends on state variables of the individual only.

The κ -rule

The energy flux mobilised from the reserve now needs to be divided over the various (lumped) processes that require these resources. Let's assume that the principal division is a binary one; a constant fraction κ is allocated to growth and somatic maintenance (the soma), and the remainder to maturity maintenance, maturation and reproduction (these processes are discussed in detail in a later section). In principle, κ is assumed to be constant over an organism's lifetime. The constant κ implies that growth and reproduction do not compete for resources directly. This is supported by the empirical fact that, in many animal species, the start of reproduction is not accompanied by a sudden change in the growth curve.⁷

Assumption: a fixed fraction of the mobilised reserve is allocated to the soma (somatic maintenance and growth), the remainder is allocated to development and reproduction.

A constant value of κ leads, together with the assumptions for feeding and somatic maintenance, to the von Bertalanffy growth pattern under constant environmental conditions. The fact that this growth pattern is generally observed in a broad range of species lends credence to the constant κ . More support for this division rule with a constant κ is provided by [177]. There might, however, be situations where the organism will deviate from this rule. One situation is (extreme) starvation. When an organism cannot pay its somatic maintenance costs, it sounds reasonable to assume that individuals would be able to decrease the investment in reproduction to fuel somatic maintenance. Furthermore, there are indications that κ is in some cases influenced by parasitic infections [86], day



⁶As a third constraint, Kooijman [155] adds partitionability, but I will not try to explain that.

⁷Notable exceptions are insects [179] and copepods [137] that have a final moult to adulthood, and do not grow anymore as adults. For these animals, the standard κ -rule cannot be used, at least not throughout the entire life cycle. This will be discussed a little further in Section 2.7.

length [265], presence of predators [173, 244] and possibly conspecifics [51]. It is almost inevitable that some man-made chemicals will act through the same mechanism, although I have not found a clear example yet.

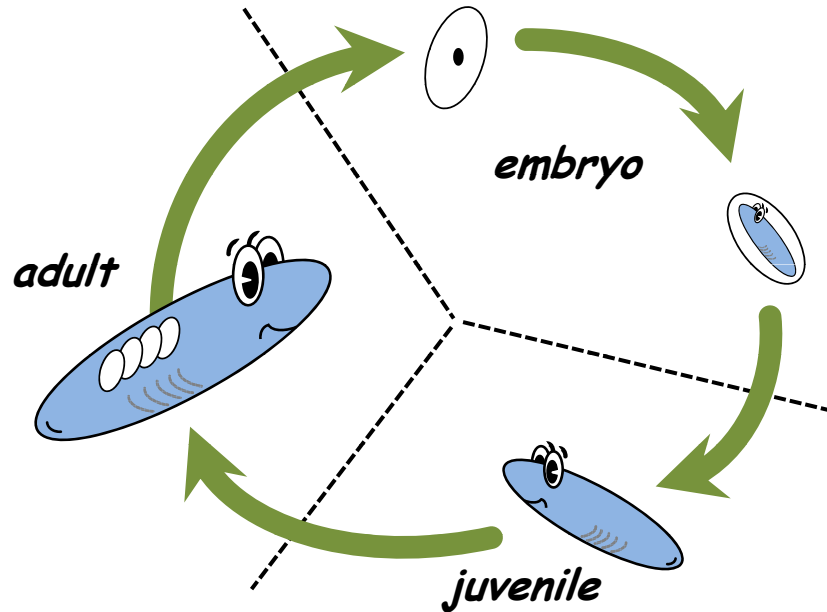


Figure 2.4: Simplified life cycle in the DEB context. The transition from embryo to juvenile is marked by the start of feeding; the transition from juvenile to adult is marked by the start of investment in gametes.

Maturation and life stages

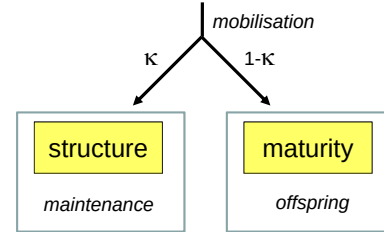
DEB theory considers three life stages: the embryo, the juvenile and the adult (see Fig. 2.4). The transition from one stage to the next constitutes a switch in metabolic behaviour. The transition from embryo to juvenile (‘birth’) is linked to the start of feeding, or at least the possibility to feed when food is available. Embryos do not feed but make use of the reserve material provided by the mother. The classification embryo or juvenile is thus made on the basis of whether or not the animal is capable of feeding. Some animals produce eggs that hatch into larvae that do not feed; such larvae should, in the DEB context, be considered ‘embryos’. The transition from juvenile to adult (‘puberty’) marks the start of investment into reproduction. This does not necessarily coincide with the appearance of the first eggs or offspring, as many species produce eggs in batches, and a range of invertebrate species hatch their eggs in a brood pouch.⁸

Moving through these three life stages, an organism needs to develop, to become more complex, which requires energy. What state variable of the organism can we use to trigger the switches between life stages? Structural size is a candidate. Many species start to reproduce at a roughly constant body size (water fleas for example). However, there are plenty of examples where a size trigger does not work. Many species reproduce long after reaching their maximum size (e.g., birds), which makes a size trigger useless. Furthermore, some stresses lead to a change in the size at puberty. In nematodes, we

⁸An example is the water flea *Daphnia magna*. The timing issues that this can cause in model analysis, when comparing model and observations, are discussed in [140].

observed a clearly decreasing size at first reproduction under toxicant stress [2, 245] (which we treated at that time with a descriptive parameter). For *Daphnia* species, length at puberty is hardly affected by food density [52, 62, 155], but it appears to be affected as a result of the presence of predators [244, 52] or infection by microsporidian parasites [47].

For the general case, we thus require a new state variable to keep track of the level of complexity, or ‘maturity’, of the animal. Becoming more complex can be viewed as learning new metabolic tricks; it requires energy, and thus has to be fuelled from the reserve, but does not lead to the build-up of body mass. In a sense it is like learning DEB theory: it costs a lot of energy, but you don’t gain weight. Thus, DEB theory keeps track of the level of complexity by the state variable ‘maturity’. The stage transitions take place at fixed levels of maturity and are assumed to be instantaneous. This maturity has the status of ‘information’; it has no mass or energy associated with it. It is quantified by the amount of reserve used to build up this level of maturity, but this energy is lost; you do not gain energy from burning maturity. Treating maturity as mass-less is advantageous: we do not have to worry about the conversion efficiency of reserve into maturity mass, and about the contribution of maturity in the physical size of the organism (which would lead to deviations from the von Bertalanffy growth curve).



The build up of maturity starts when the embryo starts to develop and stops at puberty. At that point, the investment in maturation is switched to the reproduction buffer. The concept of maturity thus allows for the constant- κ rule to work: it explains what happens in juveniles with the mobilised reserve energy in the $1 - \kappa$ branch.

Assumption: stage transitions such as birth and puberty are triggered by fixed maturity thresholds. Maturity has no mass or energy, but has the status of ‘information’, and is quantified by the amount of reserve needed to build it up. After puberty, maturity does not increase anymore.

The maturity concept is a heroic simplification of biological complexity, but it seems to serve well to explain the general patterns in stage transitions over the life cycle [17]. Whether the concept in its current form suffices to explain observed hatchling size differences (e.g., as a response to stress) remains to be more closely investigated. I will discuss an alternative in Section 2.7.

Somatic and maturity maintenance

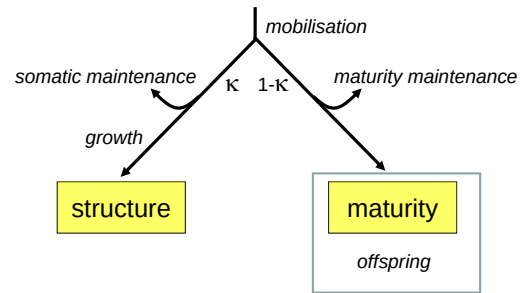
Somatic maintenance is a lump sum of all the energy-demanding processes that are needed to maintain the body’s integrity. This includes the maintenance of concentration gradients across membranes, turnover of structural proteins, and muscle activity. Reserve is assumed not to require maintenance. This is supported by the observation that freshly laid eggs, which consists almost entirely of reserve, do not use oxygen.⁹ In the standard DEB model, there is only a single structure with a constant composition,

⁹Although one could argue that the reserve material in eggs is generally quite different in composition (and purpose) from what we consider to be reserve in the later developmental stages.

and therefore the maintenance costs are taken proportional to the amount of *structural* volume. However, there might be elements of somatic maintenance that relate to surface area (osmotic work in aquatic organisms, heating in endotherms). Here, I will ignore these surface-related maintenance needs. The somatic maintenance costs need to be paid from the flux allocated to the soma first; the remainder is used to make new structure.

Assumption: somatic maintenance takes priority over growth, and maintenance costs are proportional to structural body volume (although some components may be surface related).

Similar to the somatic maintenance, there are also maintenance costs to be paid from the reserve flux allocated to maturation and reproduction. Maturity maintenance must be paid first, and the remainder of the flux can be used for maturation or reproduction. It makes sense that maintaining a certain level of complexity should require energy. Again, that can be compared to learning DEB theory: learning it takes energy, but if you do not continuously invest some energy, your knowledge will eventually dissipate. Keeping your knowledge at a constant level requires some energy for maintenance. The need for this maturity maintenance follows from the observation that many organism can be kept in the juvenile phase by keeping them at low food densities.¹⁰ If they do not die, there must be sufficient energy mobilised from the reserve to pay somatic maintenance costs. But what happens to the fraction $1 - \kappa$ that is allocated to maturation? They never reach adulthood, despite a low but continuous allocation to maturation. In DEB theory, we assume that this allocated energy is used by maturity maintenance.



Assumption: maturity maintenance takes priority over maturation and reproduction. Maturity maintenance costs are proportional to the reserve invested into maturation.

Maturity maintenance is thus a practical means to make sure that there is a non-zero minimum food level required for reaching adulthood. But what is this energy used for by the organism? In DEB theory (see [155], Section 2.5.3), maturity maintenance is “thought to relate to the maintenance of regulating mechanisms and concentration gradients” as well as the maintenance of defence systems such as the immune system, to fight diseases and parasites. These costs may be (partly) reducible under severe stress, perhaps at a cost of increased susceptibility. However, to my knowledge, there is little empirical evidence to support this interpretation.

Reproduction and the reproduction buffer

Adults do not invest in further maturation, but instead allocate this flow of energy to the production of offspring. Eggs are produced in discrete numbers, and therefore, the continuous flow of energy allocated to reproduction needs to be collected in a buffer.

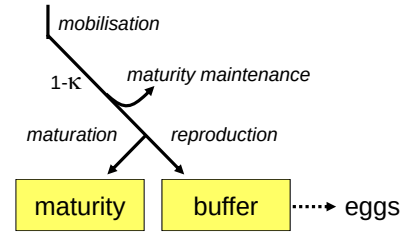
¹⁰From personal observation, I know that this can be done for the earthworm *Eisenia andrei*. When kept in high densities in laboratory culture, these worms do not mature (they do not develop a clitellum), which ensures a continuous supply of immature individuals for bioavailability and toxicity testing.

The energy in the buffer is converted into eggs at the moment of spawning. When clutch size is small, the effect of the reproduction buffer can generally be ignored. It is then possible to assume that the energy flux to reproduction is continuously converted into eggs, and we can work with a continuous reproduction rate. For population modelling, such a continuous reproduction should be considered carefully, especially at low reproduction rates. If ten mothers each have enough reproduction buffer to produce half an egg, that does not lead to the production of five offspring (see the ‘pooling effect’ in [149]).

The trigger for the conversion of the buffer into eggs is species specific. Some species may produce a single egg at the moment the buffer contains enough energy for one egg, whereas other species reproduce in clutches, perhaps only once a year, where the triggers can be internal (a value of the state variables) or external (e.g., temperature).

Assumption: at puberty, the energy flux to maturation is redirected to a reproduction buffer. At spawning events, the buffer is converted into a discrete number of eggs.

Including the dynamics of a reproduction buffer is conceptually simple, but more complex technically. At the moment of spawning, there is a discontinuity in the number of eggs and in the state of the mother (buffer contents, body weight, chemical load). Furthermore, in many experimental set-ups we do not have information from the individual mothers, and the mothers are not perfectly synchronised, which implies that it is difficult to extract buffer handling rules from the data on an experimental cohort.



Eggs consist of a tiny amount of structure (approximately zero), but a large amount of reserve. Following the strong homeostasis assumption with a single reserve compartment, reserve cannot change in composition. Therefore, the reserve in the eggs are assumed to be the same as that of the mother. The efficiency of converting the reserve in the buffer to eggs is therefore probably high, as little transformations would be required.

Assumption: initially, the developing embryo in the egg has negligible amounts of structure and maturity, but a considerable amount of reserve. The reserve in the egg has the same composition and the same dynamics as the reserve in the feeding life stages.

Maternal effects; the costs for an egg

To translate energy stored in the reproduction buffer to number of eggs, we need to know the energetic costs for a single egg. It is likely that, under optimal conditions, the mother will provide the developing embryo with sufficient resources to complete its development, and hatch with an optimal nutritional status (with the maximum reserve density for its structural size).¹¹ The standard assumption in DEB theory is that birth occurs when a maturity threshold is achieved by the embryo (as explained earlier in

¹¹I here assume, for the sake of readability, that hatching equals birth. But please note that this is not necessarily true. Birth in the DEB context implies the start of exogenous feeding, or better: the start of the *ability* to feed.

this chapter). Therefore, if we know the DEB parameters of the organism, we can use them to calculate the amount of reserve that the embryo needs to hatch with the maximum reserve density. This rests on the assumption that the DEB parameters of the embryo are the same as those of the feeding stages, and that the embryo also grows isomorphically. Furthermore, the egg-cost calculation rests on the previous assumption that hatching is triggered by a maturity threshold.

However, what is the ‘maximum reserve density’ for a hatchling? If we observe organisms in the field or in the laboratory, we can never be absolutely sure that the mothers were living under the most optimal conditions. A practical solution to deal with this problem is to assume that the reserve density at birth equals that of the mother at egg formation. In other words: well-fed mothers give rise to well-fed offspring, and poorly-fed mothers give rise to poorly-fed offspring (with low reserve density).

Assumption: the amount of reserve that mothers allocate to a single egg allows the embryo to complete its development, and to hatch with a reserve density that equals that of the mother at egg formation (a maternal effect).

An advantage of this assumption is that no additional parameters are required to estimate egg costs. Furthermore, under constant limiting food densities, the reserve density in the embryos will immediately be in steady-state with the food density in its environment, which means that the hatchling will follow the von Bertalanffy growth curve immediately after hatching. A practical disadvantage of this maternal effect assumption is that the egg costs need to be continuously calculated in environments with varying food levels, which increases calculation time. Furthermore, the experimental support for this assumption of maternal effect is limited. Some species do the exact opposite; produce larger, more starvation resistant, offspring when feeding conditions are poor (e.g., [71]). This seems to make evolutionary sense, as these offspring might stand a better chance in surviving the poor conditions. I will discuss an alternative set of rules in Section 2.7.

Mortality and the hazard rate

Empirical evidence shows that not all individuals in a test population die at the same time under the same conditions, even when great care is taken to start with a homogeneous population (even using clones). Even though it is possible that there are still differences between the individuals that determine the exact time of death, it is more intuitive to treat death as a chance process. It is unlikely that death is truly a stochastic process, but there will be so many elements that play a role that we can safely treat it as one.¹² The branch of statistics that deals with events of mortality or failure (in mechanical systems) in time is called survival analysis. This analysis makes use of the hazard rate, which is the ‘instantaneous probability to die’; or, more precisely, the hazard rate times a very small time interval gives the chance to die in that interval, given that you were alive at the start of that interval. The hazard rate will generally increase with exposure to toxicants, but will also increase with age (ageing or senescence). For independent causes of death, hazard rates may simply be added.

¹²As an analogy, throwing dice is not an inherently stochastic process. However, there are so many factors that play a role (angle of throwing, speed of throwing, rotation of the wrist, height from the table, etc.) that it can safely be treated as a chance process.

Assumption: mortality can be treated as a chance process at the level of the individual.

At this point, it is good to note that there is a separate, but related, framework for TKTD modelling that deals with effects on survival only. Focussing on survival implies that we don't necessarily need to consider the energy budget, which leads to a very simple model formulation. This GUTS framework is treated in detail in a separate e-book [118].

Parameter values as constants

Species differ mainly in their parameter values, and not so much in model structure. How are the values of the DEB parameters of an individual determined? All of the members of a species tend to have a similar life history.¹³ It thus makes sense to suggest that the value of the DEB parameters is under genetic control. In general, we can assume that the parameter values remain constant over the entire life cycle. If parameters would vary all over the place as the animal grows and develops, the model structure would be of little use. However, we know that changes in the environment (e.g., day-night cycle, temperature, toxicants) can lead to changes in the parameter values. Therefore, there is also a phenotypical aspect to the DEB parameters.

Assumption: DEB parameters are (in principle) constant over the entire life cycle.

Embryos are assumed to follow the same model structure as juveniles and adults (Fig. 2.5). The only difference being that they do not feed. Whether embryos can be assumed to grow isomorphically is not entirely clear, although the standard model provides a very good explanation for embryonic development in a broad range of animals [266]. However, the parameters for the embryo do not always match those of the feeding stages; the embryo is often slower in its development than predicted. A simple one-parameter extension was proposed, where development accelerates in an early part of the life cycle [17, 166] (see also Section 2.7).

Do embryos have the same parameter values as their mothers? In sexually-reproducing species, the offspring's genotype is of course a mix of that of the parents; the parameters of the embryo may thus differ to some extent from the mother's. In parthenogenetic species, the offspring should have the same genes as their mothers. However, there are examples where the environment of the mother affects the life history (and thus the DEB parameters) of the offspring (see [33] for review, and [220] for a DEB analysis). The DEB parameters will thus be determined by the combined influence of genes and environment.

2.3 Summary and evaluation of the standard DEB model

The standard DEB animal model follows from a series of assumptions (slightly modified from [155]):

1. The basic state variables of the individual are reserve, structure and maturity. Reserve and structure have a constant composition (strong homeostasis), and maturity represents information (and therefore has no contribution to overall size or mass).

¹³For species with sex differentiation, the sexes may differ in their traits and thus in their DEB parameters.

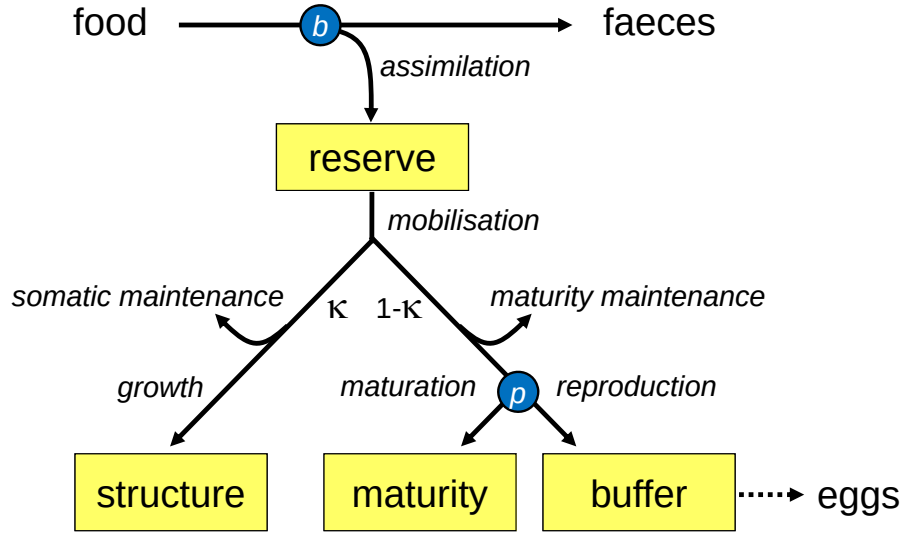


Figure 2.5: Schematic diagram of the energy flows in a standard DEB animal. The nodes b and p denote switches at birth (start of feeding) and puberty (start of reproductive investment). The mobilisation flux is split according to a constant fraction κ .

2. When maturity exceeds a threshold value, the individual starts feeding ('birth'). A higher maturity threshold marks the start of reproductive investment ('puberty'). Above the puberty threshold, maturity does not increase any further.
3. Food is instantaneously converted into reserve, which in turn fuels all metabolic processes. The mobilisation rate of the reserve depends on the value of state variables of the individual only.
4. The embryonic stage initially has a negligible amount of structure and maturity, but a substantial amount of reserve. The reserve in the egg has the same composition and the same dynamics as the reserve in the feeding life stages. The reserve density at birth equals that of the mother at egg formation (maternal effect). The developing embryo has the same value for the DEB parameters as the mother.
5. The feeding rate is proportional to the surface area of the individual, and the food handling time is independent of food density. Assimilation efficiency is constant and independent of body size or food density. The total time budget consists of the sequential processes of searching for food and handling it.
6. At constant food density, the ratio between the amount of reserve and structure becomes constant (weak homeostasis).
7. Somatic maintenance is proportional to structural volume, but some components may depend on surface area (e.g., osmotic work in aquatic organisms and heating in endotherms).
8. Maturity maintenance is proportional to the level of maturity.
9. A fixed fraction of the mobilised reserve is allocated to somatic maintenance plus growth, the rest to maturity maintenance plus maturation or reproduction (the

κ -rule). Somatic maintenance takes preference over growth, and maturity maintenance takes preference over maturation or reproduction.

10. The resources allocated towards growth are converted into structural biomass with a fixed conversion factor. The resources allocated towards reproduction are converted into eggs with a fixed conversion factor. Buffer-handling rules can be used to convert the continuous allocation to reproduction into discrete batches of eggs.
11. The individual does not change in shape during growth (isomorphism).
12. Mortality can be treated as a chance process at the level of the individual.
13. The DEB parameters are (in principle) constant over the entire life cycle.

These assumptions specify the standard animal model. This specification is internally consistent and unique: from these assumptions, in principle, only one single set of model equations is possible. The standard model specifies the patterns of feeding, respiration, growth and reproduction over the entire life cycle of an individual animal (including the embryonic stage), using a limited set of model parameters. Even though we restricted ourselves to animals, the enormous diversity in life cycles and physiology implies that this simple model must be wrong in detail. Obviously it is. Some of the assumptions underlying the standard model can be relaxed without threatening the core of the theory. For example, the standard model can readily be modified to accommodate changes in shape (Assumption 11), and for some species (or some life stages of specific species), feeding should be related to volume rather than surface area (Assumption 5). Some life cycles are better captured assuming a switch in parameter value at some life stage, e.g., at metamorphosis (Assumption 13). Within Assumption 4, the maternal effect rule can be modified without major consequences for the model. Some of these modifications are discussed in Section 2.7.

The other assumptions can be considered to lie at the core of DEB theory. It is quite hard to come up with a different, similarly sized, set of assumptions while maintaining the same power of explanation that DEB theory has demonstrated over the years. The list of assumptions needs to be internally consistent, and consistent with the laws of physics and well established empirical patterns. It may be hard to come up with alternatives, but is it impossible? To support the standard DEB model, a set of ‘stylised facts’ has been presented [240, 241]: “empirical patterns that are the ultimate test for any metabolic theory.” Since these papers are presentations of the theory, it is perhaps not surprising that DEB is found to be consistent with all of these ‘facts’. In a subsequent paper, Lika and Kooijman [177] explored a wide range of possible topologies for energy budgets and concluded that the standard DEB model is the only one to pass all these tests. Kooijman later even went one step further and claimed that no alternative exists, nor will ever exist [158]. I, however, believe that these claims are premature, and may even stifle further research and critical questioning of the theory. The crux of my argument lies with the support for the ‘stylised facts’: this list of facts indeed comprises patterns that are produced by the standard DEB model, but to what extent are they well-established empirical patterns?

In 2012, I began to explore the applicability of a reserve-less model (DEBkiss, [132, 109], see Section 2.6), which applies a slightly different, but still closely-related, set of

assumptions. In this model, the resources obtained from feeding are directly used for growth, maturation and reproduction, without a reserve compartment (though there can be storage as a reproduction buffer in the $1 - \kappa$ branch, and as yolk buffer for the embryo). Since DEBkiss lacks a reserve compartment in between assimilation and the downstream processes, it should be clear that this model will fail some of the tests embodied by the ‘stylised facts’ (see [177]). Therefore, I set out to scrutinise the references brought in to support the ‘facts’ that are inconsistent with DEBkiss. I present a detailed paper-by-paper analysis in the DEBkiss e-book ([109], Appendix B). My sobering conclusion is that many of the ‘stylised facts’, brought in to support the reserve compartment and its specific dynamics, are poorly supported by the references provided. In some cases, the patterns are simply not there, in others the observations can be interpreted in various ways. As an example of the latter, structural growth continuing in the absence of food would point at the presence of a DEB-style reserve. However, four of the studies cited to support this fact deal with continued growth in early-life stages, where yolk was reported to be present. No one will question the fact that an animal can grow on its remaining yolk in the absence of food. However, should this yolk be treated as part of a DEB-style reserve? The reserve-less DEBkiss treats yolk as a storage buffer, handed over by the mother, that is assimilated by the embryo. DEBkiss therefore also predicts continued growth in the absence of food as long as yolk is present.¹⁴ Therefore, such studies cannot be used to defend the DEB-style reserve; the interesting question is whether (and to what extent) individuals *without* yolk can grow (in structure) on their DEB reserve only.

Apparently, it is difficult to find data for animals that unequivocally point at the presence of a reserve compartment with the specific DEB dynamics. This also links to the experience so far with DEBkiss: this simplified model without reserve provides an excellent explanation of observed life histories for a wide range of species and environmental conditions.¹⁵ Also in a recent ecotoxicological case study, DEBkiss and DEB performed virtually identically in a direct comparison within a risk-assessment context [227]. This is not to say that animals do not have reserve; I am questioning whether that reserve is important enough to justify its own dynamic compartment (for all animal species), and whether that dynamics should then follow the DEB rules.

In view of the above, more modesty is warranted: the standard DEB model has a fantastic track record, but it is too early to conclude that no useful alternatives can exist (and too early to discard DEBkiss as a useful alternative). First, we need to prune the list of stylised facts: these need to be well-established empirical patterns that can indeed be used as ultimate tests for energy-budget models. Presenting predicted patterns from the standard DEB model as empirical facts is misleading. Secondly, we need better data sets for testing these models. It is stunning that, with all of the data-collecting going on (see [192]), there is so little unequivocal empirical support for (or against) some of the patterns predicted by DEB. We need dedicated and detailed experimental work, designed specifically to test the predictions from energy-budget models; collecting what has been gathered in other contexts (e.g., data on field-collected animals) simply does not suffice. Given the enormous range of successful applications of the standard model, I would not be surprised if the majority of the assumptions would survive such closer

¹⁴A similar buffer was proposed in the reserve-less net-production model of [178], which therefore can also reproduce this fact. In the comparison of topologies for energy-budget models [177], such buffers were not considered.

¹⁵See list of paper at https://www.debttox.info/debkiss_appl.html and the dedicated e-book [109].

scrutiny. Nevertheless, theories benefit most from being challenged, and, by critically looking closer, we might learn where the limits of DEB theory (and the simplified frameworks) lie.

2.4 Response to different environments

Comparing different constant food densities

The DEB assumptions state that food uptake is proportional to a surface area, and maintenance costs are proportional to structural volume. When the animal grows isomorphically, the volume will increase faster than the surface area. Therefore, at a constant food density, growth will slow down and stop when the entire energy flux allocated to the soma (for structural growth and somatic maintenance) is needed to pay the somatic maintenance costs. At constant low (limiting) food densities, less energy is coming in, which implies that growth ceases at a smaller body size; the decreased energy intake cannot support the maintenance costs of a large structural volume. Less energy intake also means less investment in maturation and reproduction, generally leading to a delay in the start of reproduction and a decreased maximum reproduction rate, compared to high food levels. Below a certain critical food density, the organism will not have the possibility to mature anymore.

Experiments at various constant food levels are valuable to estimate the model parameters, especially those that specify reserve dynamics (see Section 7.2). As long as food is provided *ad libitum*, we do not need to consider the actual food density in detail; the animal is feeding at the maximum rate for its size anyway. At lower food levels, however, keeping food levels constant in an experimental system is often more complicated than it may seem. One problem is that the animals remove food particles from their environment, and thereby decrease their own food density (in a size-specific manner). Keeping food density constant and limiting for aquatic organisms is best done in flow-through setups, as long as the animals eat small particles floating in the water (like *Daphnia* feeding on algae). However, when feeding on larger food items, the concept of ‘food density’ may not even apply. For example, pond snails (*L. stagnalis*) are usually fed pieces of lettuce, and rats are fed pellets. We can feed these animals less than *ad libitum*, but it would be hard to ensure a *constant* degree of food limitation when the animals grow.¹⁶ Some practical approaches that can be explored are to dilute the food with an indigestible component, applying a regime alternating *ad libitum* feeding with starvation (e.g., [87]), or to measure actual feeding or defecation rates.

Varying food densities; starvation

Because animals have a reserve compartment, the energy-consuming processes such as growth, maintenance and reproduction are to some extent buffered from the actual food availability. As long as the animal is still growing, it will respond to a decrease in food availability by a decrease in the growth and maturation/reproduction rates. When the reserve density decreases, the mobilisation of energy from the reserve decreases. First, the maintenance costs need to be paid, and the remainder can then be used to

¹⁶For a mammal example, in [99], the rats received a limiting daily food amount as a percentage of the *ad libitum* requirements of the adults, throughout their life time. For juveniles, this represents much less of a food limitation than for the larger-sized adults.

grow and mature or reproduce. However, after more severe starvation, the reserve will decrease to a point where the κ -flux of the mobilised reserve is not sufficient to pay the maintenance costs anymore. For fully-grown animals, the problems are more acute. For these animals, all of the mobilised energy that is allocated to growth and somatic maintenance is already used for maintenance. A very small decrease in reserve density would thus immediately lead to problems (this is discussed in detail in [250]). In reality, fully-grown animals do not die immediately when they experience a decrease in the food density, so it is clear that organisms have a strategy to deal with starvation.

Reducing somatic maintenance costs is probably not so easy for most animals, and generally requires some major adaptations. Some mammals can drop their metabolic rate as they hibernate during the cold winter months, whereas other animals do the same when they aestivate during the dry summer periods (e.g., earthworms, land snails, reptiles and some land-dwelling arthropods). Marine zooplankton species such as copepods and krill can reduce their metabolic rates during the winter season when food is scarce [96, 248]. Hummingbirds drop their body temperature and metabolic rate every night (they enter torpor), as they cannot maintain their high-octane lifestyle without regular refuelling. Bats do the same thing, but during the day. Other animals can enter a dedicated resting stage, often exclusively in a certain stage of their life cycle. For example, many opportunistic soil nematodes (e.g., in the family Rhabditidae) can enter a metabolically suppressed state (called a ‘dauer larva’) under harsh conditions, but only in a particular larval stage. The water flea *D. magna* can produce resting eggs which can survive long periods of adverse conditions. When an animal cannot evoke such specialistic adaptations to decrease its maintenance costs, it either has to die or deviate from the standard DEB rules. It is very likely that they choose the latter option, but it appears that they do this in a way that varies a lot between species. Strategies could include the following aspects:

1. The reproduction buffer is made up of reserve, and may be used to pay somatic maintenance costs. Adults may stop conversion of the buffer into eggs, and keep the buffer to survive under starvation. Juveniles cannot use this strategy as they do not have a reproduction buffer; the maturation that has built up cannot be used to fuel maintenance costs as it is thought to represent ‘information’. However, juveniles might be able to tap into the energy flow directed towards maturation and use this for somatic maintenance instead. In general, it is thus possible that animals pay somatic maintenance costs under starvation by taking what is needed from the $1 - \kappa$ flux.
2. Change κ to redirect enough energy from the flow going to maturity maintenance and maturation/reproduction to make sure that somatic maintenance can be paid. Organisms might change κ gradually to just match the maintenance needs (which would be very similar to the previous strategy). Changes in κ may not be unrealistic, since this parameter is probably under metabolic control: in pond snails it seems to change with day length [265], and in daphnids with the presence of predators [244].
3. Another strategy may be to change the mobilisation of the reserve: withdraw from the reserve the energy needed to pay somatic and maturity maintenance (and stop all investment to maturation and reproduction). Given the proposed

surface-related mechanism behind reserve dynamics [187], the step to change the dynamics in such an *ad hoc* fashion should not be taken lightly.

4. Reduce maturity maintenance. In DEB theory, maturity maintenance is thought to represent a.o., the immune system. To some extent, these costs may be reducible under stress, although there will be repercussions. Perhaps, maturity decreases when maturity maintenance is not paid (which leads to rejuvenation), and perhaps the resistance towards diseases and infections is reduced.
5. Use structure to pay maintenance costs. Organisms are probably able to burn at least some part of their structure to pay their somatic maintenance costs. However, the energy needed to make a unit of structure cannot be returned fully when this unit of structure is used for maintenance. Further, there is probably a (species-specific) limit to the factor of shrinkage of structure that an organism can survive (and possibly no full recovery on re-feeding).
6. Some species may channel all of the possible energy that can be mobilised (perhaps including a part of structure) towards reproduction, after which the animal dies (emergency reproduction).

The strategy that a particular (life stage within a) species follows is probably very specific to that species. Furthermore, species may use more than one of the elements outlined above in their strategy. Therefore, it is difficult to come up with a general approach.¹⁷ The strategies listed above have a certain elegance as they do not require large changes in the metabolic organisation, and do not necessarily require additional parameters. However, when you want to analyse a situation where your animal experiences varying food densities (or a stress on feeding or assimilation) you should take care to work out an appropriate strategy. Despite the degrees of freedom offered by DEB, there is still a very stringent restriction, and that is that your solution must be consistent within the theory and thus obey the conservation laws for mass and energy. This is an area of the application of DEB theory where more dedicated research is urgently needed, especially as the choice of starvation strategy can dominate population dynamics [195], and because exposure to toxicants may induce starvation effects when it affects assimilation or maintenance (even under *ad libitum* food availability, see [107]).

Changes in temperature

Especially for ectotherms, a change in temperature has drastic consequences for the life history. All metabolic processes increase with an increase in temperature, up to a certain point. The most logical starting point to deal with temperature effects is to assume that all rate constants are affected by temperature by the same factor (see [155], Fig. 1.7). If this factor would be different for the various metabolic processes, the organism would have a hard time maintaining its normal operation. The Arrhenius relationship usually describes the temperature dependence of metabolic rates well, at least within a ‘relevant’ range of temperatures for the organism (it will fail at the extremes, where temperature becomes a stressor rather than an environmental factor).¹⁸ This relationship contains only a single parameter (the Arrhenius temperature).

¹⁷The updated DEB-based TKTD models [107, 125] follow strategy 1 and 5. First, resources from the $1 - \kappa$ flux are used for somatic maintenance, but if this is insufficient, structure is used and the animal shrinks.

¹⁸A proposal to modify an effect model for survival (GUTS) to include thermal stress was presented in [188].

More experimental work is needed to evaluate this simple model extension. In copepods, for example, lowering temperature increases ultimate body size [137]. In springtails, the opposite trend was observed [87]. This indicates that not all rate constants are changing with temperature by the same factor, and that the deviations may be species specific. Furthermore, work is needed to investigate how toxicity-related parameters change with temperature. For the midge, *Chironomus riparius*, Koch *et al.* [152] found that the Arrhenius correction for all DEB rate constant successfully explained the growth curves at four temperatures. However, for the toxicological rate constants, the authors could not use the same factor. Also with survival models, studies indicate that the consequences of temperature changes on toxicity parameters may not be so straightforward to predict [94, 189].

2.5 Link to measurable properties

The standard DEB model specifies structure and reserve over the life cycle. However, these properties cannot be observed directly. Furthermore, many model parameters have ‘energy’ in their dimension and thus cannot be estimated without some energy-based measurement (which is rather uncommon practice in ecotoxicology). However, the observations that are usually made (length, weight, numbers of eggs, number of survivors) have obvious links to the model entities. This is the area of auxiliary theory, which provides the link between the idealised theoretical world and the dirty real world (see also [157]).

Primary and compound parameters

The primary parameters of DEB are directly linked to metabolic processes, such as the energetic costs in Joule for a unit of structure. Determining the value of a parameter with a unit that include Joules requires a measurement that includes Joules. This is problematic when we want to estimate parameters based on observations on length, weight and numbers of offspring. We can circumvent this by using fixed values for conversions, e.g., for the energetic content of structure and reserve (see [176]). However, we can estimate the ratio of two parameters that both have energy in their dimensions. DEB models can be formulated in terms of such compound parameters such as the ‘energy investment ratio’ (the ratio of the growth costs over the maximum potentially available energy for the soma). The use of compound parameters ensures that we can estimate them from the available data, but hinders the interpretation of their values. Further, we have to take care to correctly include toxicant effects on the primary parameters into the compound ones. More information about primary and compound parameters can be found in [167].

Structural length and real length or weight

As long as the organism grows more-or-less isomorphically (does not change in shape), we can, in principle, use any length measure as a proxy for structural length. Every well-chosen length measure will be proportional to the structural length. However, one does have to watch out not to include body parts that do not grow in proportion to the whole body (e.g., the spine in daphnids, or beaks of birds), or length measures

that are influenced by the reserve (e.g., belly circumference in humans). The exact details of this proportionality do not matter for each application, and in ecotoxicology we can usually ignore these details. The value of any parameter that has body size in its dimension is susceptible to the actual length measure that is used. If we want to compare such parameters between species, we need to think more carefully about the conversion to a the ‘real’ (volumetric) structural length (see [155] for details).

If we measure weight or volume of the organism, we have to realise that these measurements are a weighted sum of structure, reserves and reproduction buffer (and possibly also the associated products and the gut contents). If we can assume that the reserve density is constant (constant food density, and no toxicant effects on feeding or assimilation), the reserve mass is a constant proportion of structural mass. If we can furthermore ignore the contribution of the reproduction buffer, the cubic root of body weight is proportional to the structural length.

In practice, most ecotoxicological applications of DEB have ignored this problem altogether and used any physical length measure or the cubic root of volume or weight as proxy for structural length. Clearly, this is not correct in detail, but quantitatively the consequences may be minor in many cases. Some species, however, have a metamorphosis in their life history (e.g., insects and many marine invertebrates, see Section 2.7), which can involve a huge difference in shape. In such cases, the relationship between physical length and structural length requires more attention.

Hazard rates to survival

DEB theory specifies hazard rates, and thereby survival probabilities. These probabilities cannot be observed directly, but have to be inferred from survival patterns in populations with, preferably, a large number of individuals. This is dealt with in Chapter 7.

2.6 Which model to use from the theory?

DEB is a theory that offers a framework for understanding how organisms obtain resources and how they allocate them over the various sinks during their life cycle. In practice, you will work with specific models that are based on this framework. These models can be extremely simple or extremely complex, depending on the question that needs to be addressed and the amount of information that is available. In ecotoxicology, the data sets are usually so limited that stringent model simplification is extremely helpful. Figure 2.6 provides a schematic overview of various DEB models that have been, and are being, used for addressing ecotoxicological questions (and stress ecology in general). These models are placed in an axis framework from simplified to full, and from primary parameters to compound parameters. Simplified models, as the name implies, sacrifice some of the ‘details’ of DEB theory; in general, they can be viewed as special cases of the standard DEB animal model. Using compound parameters rather than primary parameters does not necessarily imply a model simplification; it can often be done while keeping the model structurally identical. These two axes also form the basis of the proposal for naming model variants by Sherborne *et al.* [236] (which I do not strictly follow here).

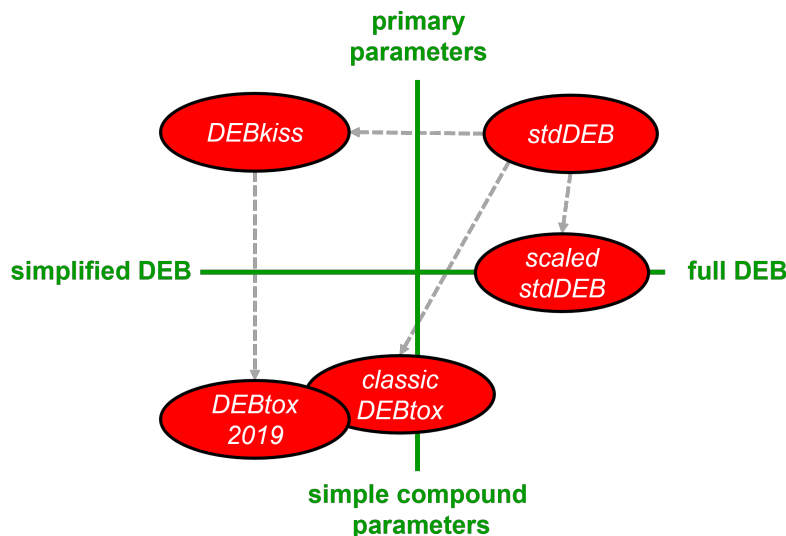


Figure 2.6: Placing various specific DEB models in an axes system with the axes primary to compound parameters, and simplified to full. Arrows indicate how the versions, in general terms, relate to each other (although the complete historical background is somewhat more complex).

The standard DEB animal model

we start in the top-right corner of Figure 2.6 with the standard DEB animal model and its extensions (here referred to as *stdDEB*). Even though it is a relatively complex model, in primary parameters, this model has been used to analyse ecotoxicological data sets [16, 264]. It was also proposed for application in risk assessment by Baas *et al.* [19]. In a more recent publication [125], the *stdDEB* model was combined with an up-to-date TKTD module, and demonstrated in a case study. This model is too complex to be fully parametrised from ecotoxicological data sets alone. However, the add-my-pet library can help here [192]: this library contains basic parameter sets for a wide range of species, including many of the species that are commonly used in ecotoxicology. This library helps practical use of *stdDEB* considerably, without needing to resort to simplifications. However, a number of issues remain. Firstly, the quality of the entries in the add-my-pet library differs considerably between species. Not many species have the underlying data sets that suffice to confidently parametrise the model (defaults or links to related species would then be used to fill the data gaps). Additionally, various data sets for the same species may be inconsistent. In many cases, data sets are combined that originate from different laboratories, using different strains of the species, different food, different (or poorly described) conditions, etc. Each entry in the library is thus a compromise between various information sources. Related to this, it is unlikely that the entry will actually provide an accurate description of the control in a specific toxicity test. A close correspondence is needed for the model to be able to identify toxic effects as deviations from the modelled control situation. One or more of the basic DEB parameters will thus have to be re-fitted for each toxicity data set. At this moment, it is not so clear which parameters to select and how that should be done (see [125]).

The scaled standard DEB model

In the types of questions commonly encountered in ecotoxicology, ‘energy’ does not play a role explicitly. We do not measure energies, and therefore have no possibility of directly determining parameters with energy in their dimension. In those cases, it makes sense to work with a DEB model where the energies are scaled out. As described in [167] and the technical background document, an efficient scaling is to divide reserve energy by the surface-area-specific assimilation rate. The result is a full-scale DEB model without the dimension ‘energy’; the only remaining dimensions are time, length and numbers (for reproduction). Fitting the model to typical ecotox data sets is however not so simple, and the resulting parameters are difficult to interpret. This model was used by me and my co-authors in several publications from 2010-2011 [139, 141, 129], as well as in the population model DEB-IBM [197, 196, 195]. This model formulation seems to have fallen out of grace for analysis of ecotoxicity data, though: the initiation and expansion of the add-my-pet library allows use of the unscaled model, while the seekers of simplicity have moved towards further-simplified model versions.

The classic DEBtox model

An appropriate simplification for standard ecotoxicity tests was presented by Kooijman and Bedaux in 1996 [161] to analyse the results from 21-day *Daphnia* reproduction tests.¹⁹ This simplification is based on the following assumptions:

Assumption: length at puberty is constant; size is a perfect proxy for maturity. Each egg constitutes the same amount of resources, and this amount is not affected by the reserve status of the mother. However, the total egg costs for the mother can be affected by a direct toxic stress on the overhead costs. The reserve is always in steady state with the food density in the environment. Maturity maintenance has a particular value, which, in stdDEB, would lead to a constant structural size at puberty across different food levels.

With these assumptions, we can remove maturity as a state variable entirely, and circumvent the rather nasty calculations required for the maternal effect. The egg costs are hidden in a compound parameter (the maximum reproduction rate), and thus not linked to initial size of the animals anymore. Additionally, we assume instantaneous steady state of the reserve, so we can remove the reserve as a state variable. The resulting model is phrased in terms of easy-to-understand compound parameters, such as maximum size and maximum reproduction rate, instead of the more abstract parameters of the full and scaled models, discussed above. This model simplification is a useful starting point for analysing toxicity data, as long as we can ignore the dynamics of the reserve and as long as there are no (large) changes in the size at puberty. However, the simplifying assumptions restrict us in the DEB parameters that can be affected by toxicants. For example, we cannot change κ in this model because a change in this parameter likely affects the size at puberty, and it is hard to include a consistent starvation module.

¹⁹This model is often referred to as the ‘DEBtox’ model, after the software that was released in 1996 to analyse some standard ecotoxicity test results [159]. Since then, this term has often been used more broadly to encompass all use of DEB models to address (eco)toxicological questions. Sherborne *et al.* [236] proposed the term DEB-TKTD for the general case. I now prefer to restrict the term DEBtox for simplified models, but it is good to be aware that different naming conventions are used.

The original derivation of the simplified equations by Kooijman and Bedaux [161] in 1996 contained a few errors. Most of these were repaired by Billoir and co-workers [37] in 2008, but a full new derivation was presented in [142] in 2012. The new derivation also allows to loosen one of the simplifying assumptions: it is possible to include reserve dynamics without adding parameters (it only requires an additional state variable). Especially when there are rapid changes in food level, or when there are rapid stress effects on feeding and assimilation, reserve dynamics could play a role. We can take the simplification even further when we are interested in growth only, under constant conditions. In this case, the standard DEB model reduces to the von Bertalanffy equation with three parameters (initial size, growth rate constant and maximum size). This simplification was proposed for the analysis of toxicity data for fish growth [160].

At this moment, the classic DEBtox model is not used much anymore. In recent years, most ecotoxicological applications have either used the standard DEB animal model or the updated simplified DEBtox2019 (discussed later in this section).

The DEBkiss model

In 2013, simplification of DEB animal models took a new turn with the publication of DEBkiss [132, 109], where the acronym KISS stands for “keep it simple, stupid.”²⁰ It follows from the same assumptions as specified for the classic DEBtox models above, but fully removes the reserve: assimilates obtained from feeding are directly used for all other metabolic processes.

Assumption: length at puberty is constant; size is a perfect proxy for maturity. Each egg constitutes the same amount of resources, and this amount is not affected by the status of the mother. However, the total egg costs for the mother can be affected by a direct toxic stress on the overhead costs. There is no reserve compartment: assimilated resources are directly used for energy-requiring processes. Maturity maintenance has a particular value, which, in stdDEB, would lead to a constant structural size at puberty across different food levels. Embryos assimilate a buffer of resources provided by the mother in the egg (e.g., yolk), and are born when the buffer is empty.

This allows for a model that is easier to understand, easier to apply, and has a closer link to measurable properties. DEBkiss includes the embryonic phase, and has an explicit mass balance with direct access to all mass fluxes, using the primary parameters instead of compound ones. This makes it possible to include additional model outputs such as respiration, and include consistent rules for the energy allocation under starvation. Furthermore, access to the mass fluxes aids the consistent application of stress factors on the primary parameters (as discussed in more detail in Chapter 3).

In contrast to DEB, DEBkiss does not aspire to provide an over-arching theory for all of life. Instead, it aims to present a framework for developing models for practical applications where simplicity, transparency, and ease-of-use are crucial (even at the expense of some biological realism and generality).²¹ In DEB theory, the reserve density

²⁰See for more information on the KISS principle https://en.wikipedia.org/wiki/KISS_principle. For more information on DEBkiss, see <http://www.debttox.info/debkiss.html> and the free e-book [109], where the model is explained in detail.

²¹In fact, I originally developed this framework as a teaching tool, to introduce DEB-TKTD concepts in a more palatable form. Its excellent performance (particularly for invertebrates) later led me to prefer the DEBkiss framework for many practical applications.

is expected to decrease with decreasing body size. This would make a reserve-less model most suitable for small (invertebrate) animals. Indeed, this is where most applications have so far focussed on; it remains to be seen which species challenge the limits of DEBkiss.

I believe that the DEBkiss framework offers particular potential for applications in ecotoxicology and environmental risk assessment, at least for invertebrate species and fish. For such applications, a formulation in compound parameters is usually most practical, which is the topic of the next paragraph.

The DEBtox2019 model

The DEBkiss model can be easily rewritten into a model version with easy-to-understand compound parameters. A complete model was presented by me in 2019, though the publication appeared in 2020 [107]. I view this as an update of the classical DEBtox model, hence the name. This model was presented with an up-to-date TKTD module (explicitly considering damage) and a consistent starvation module (since starvation may easily occur for animals under pulsed exposure to toxicants affecting assimilation or maintenance processes). The formulation in compound parameters makes the model less flexible: some mechanisms of action cannot be implemented, and it is harder (sometimes even impossible) to extend the model to include additional endpoints (e.g., respiration). Nevertheless, this is a simple-to-use and transparent model version, which can generally be completely parametrised from the data available in ecotoxicological tests (where only growth and reproduction are followed).

At this moment, I consider this model to be the best place to start for ecotoxicological questions, and particularly for routine use in risk assessment. The basis of the the DEBtox2019 model bears a close relationship to the Kooijman-Metz model of 1984 [165]. So, in a sense, we are returning to the roots (although the DEBtox2019 model makes important advances compared to its ‘ancient’ predecessor). The DEBtox2019 model is demonstrated in the case studies of Chapter 8.

Survival models

The most radical simplification can be used for short term effects on survival (ignoring growth and reproduction entirely); the DEB framework reduces to the hazard rate [30]. This reduction is so radical that it would be misleading to refer to such survival models as DEB models as all aspects of the energy budget are removed. Since 2011, hazard-based survival models are considered as special cases of the overarching GUTS framework [113], which unifies virtually all published TKTD approaches for survival.²² It should be noted that when the organisms grow or shrink, or when they reproduce, a less simplified model, including DEB aspects, might be needed to properly assess toxicokinetics (and damage dynamics), even when the focus lies on survival effects.

2.7 Deviations and extensions

The standard model is a good starting point for many questions related to stress in animals. In this section I will present some deviations and extensions of the standard

²²For more information, see http://www.debttox.info/about_guts.html and the free e-book [118].

model that are worth considering. However, we do need to be careful when introducing *ad hoc* model changes: all organisms share an evolutionary tree, so we should consider these adaptations in light of their relationships to the metabolic structure of related species.

Alternative hatching rules

Every model is a simplification of reality, and is thus wrong. Most of the assumptions behind the standard model are defensible simplifications, in my opinion. Nevertheless, none of these rules are written in stone. Here, I would like to discuss the assumptions regarding the energy investment into eggs (the maternal effect) and the trigger for hatching (at a fixed maturity threshold). In DEB theory these two aspects are tightly linked: the investment per egg is set in such a way that the embryo will hit a maturity threshold, and is born, at the point where its reserve density equals that of the mother at egg formation. The reason for a closer look at these assumptions is twofold: the empirical support is rather limited, and the choice of assumptions can have far-reaching consequences. To elaborate on the latter, an appropriate interpretation of the hatching rules and maternal effects is needed to correctly interpret the reproduction rate. Because we are focussing on mass and energy balancing, we need to know the investment per egg to interpret the number of offspring and how this endpoint responds to a stressor. Incorrect assumptions for the egg costs may thus create bias in the parameter estimation and interpretation of stress effects on numbers of offspring. Furthermore, the calculations involved with this assumption are awkward, especially under toxicant stress, as we need to predict development of the embryo to set the correct egg weight. This also raises the question of how the poor mother should make this complex prediction.

In standard DEB, well-fed mothers give birth to well-fed offspring, and thus invest more energy per egg than poorly-fed mothers. Is this prediction supported by empirical evidence? There is a lot of literature, especially on marine invertebrates, about egg sizes and its ecological and evolutionary significance. The reviews of [34, 201] show that egg size is highly variable between related species, within species, within individuals, and even within a single clutch of eggs. Egg size is influenced by the mother's status (her age, size, feeding status, etc.) and the environmental temperature. The direction of these effects also varies between species. For example, in some species, there is no effect of maternal feeding status on egg size, sometimes it follows the DEB predictions, and in others it goes into the opposite direction (smaller eggs for well-fed mothers, e.g., [71]). One problem is of course, that egg size does not necessarily correspond to energetic investment. However, this large variation in egg size, and the diversity of maternal effects on egg size, do not lend credence to the DEB assumptions.

The next thing to look at is the fixed maturity level at birth. I think there is a good case for linking maturity levels to developmental stages (see [17]), and therefore, it is reasonable to assume a maturity threshold for puberty. There must also be a minimum level of complexity before an organism can feed on its own, but does that mean that birth *has* to occur there? Some interesting clues can be obtained from experimental manipulation with embryos [239]. Two species of sea urchin within the same genus (*Strongylocentrotus*) differ in egg volume by a factor of six. The species with smaller eggs produces smaller and simpler larvae. When, however, the eggs of the species with the large eggs are reduced in size at the two or four-cell stage, fully functional larvae

result, which are smaller and simpler in form. When the eggs are reduced to the egg size of the species with the small eggs, the development in both species was very similar indeed. In all treatments, there was no effect on size at metamorphosis (when they turn into the familiarly-shaped urchin). In these species, it seems that birth is triggered by the reserves running out, and that the maturity level at birth varies. The investment per egg may be rather flexible (subject to evolution and environmental cues) to produce either a lot of small simple larvae, or a few large and complex ones.

Let's look more closely at a particularly popular test species in ecotoxicology: the water fleas of the genus *Daphnia*. These species start to reproduce at roughly the same length at different food levels [62]. This suggests that structure is a good proxy for maturity, and thus that structural size of the offspring at birth should also be unaffected by food availability. The maternal effects rule prescribes that, at low food density, mothers invest less per offspring, such that they are born with a lower reserve density than their conspecifics at high food levels. However, these predictions do not hold. At low food densities, larger eggs or offspring are produced [79, 85, 53, 52, 71], although the response seems to vary between clones of the same species [78]. This difference in size is unlikely to be explained away by differences in water content, as the eggs from the brood pouch had more dry weight [78], more lipids and protein [85], the larger offspring also turn out to be more resistant to starvation [79] (although this was not observed in [78]), and reached maturity earlier [52]. Furthermore, offspring size or 'quality' seems to increase with the size or age of the mother [79, 78, 61, 88, 52, 71] or presence of an insect predator [52], and decreases with crowding [51]. Another interesting clue comes from the observation that exposure of mothers to a certain chemical leads to the production of more but smaller offspring [88]. These facts indicate that there is a lot of variability in the investment per egg, based on the mother's status or the environment it experiences. Apparently, the embryo is able to deal with this variation in supplied reserve materials. These observations are clearly inconsistent with the current set of rules in DEB (Assumptions 2 and 4 in Section 2.3).

As an alternative set of rules, I would like to suggest the cost of an egg as a primary parameter. The mother 'decides' how much reserve she will allocate to each egg. This parameter may vary with age, size or reserve status of the mother (in a species-specific manner), and may be affected by toxicants. The embryo develops inside the egg and hatches when reserve density hits a certain level (this makes maturity at birth a model output rather than a primary parameter as in the standard model).²³ I suggest to take the 'optimal' reserve density as the mark for birth; the reserve density of the mother under *ad libitum* food. However, this has a problem: how do we know that the mother has been fed *ad libitum*? Perhaps another food source would have been more optimal. Clearly, dedicated research is needed to decide upon the most useful general rule in DEB theory. It is also possible that different groups of animals require a different rule. In any case, it is advisable to examine the size or weight of eggs and/or hatchlings in experiments, and be especially watchful when these change with stress treatments (e.g., [88]). I hope that this discussion shows that the DEB assumptions should not be followed dogmatically: they can (and should) be questioned, and alternatives tested. This particular alternative leaves the core of the DEB model intact.

²³ A similar rule is applied in DEBKiss [132]. In this model, there is no 'reserve' in the DEB sense but a storage buffer in the egg which can be assimilated by the embryo. Birth occurs when the buffer runs out. A detailed application of this model can be found in [133].

Deviating development in early life stages

Even though the standard DEB model often fits the embryonic development well, the resulting parameters do not always match those of the juvenile/adult stages. In a number of cases, the embryo (or the early juvenile) develops slower than predicted from the later part of the life cycle, pointing at some form of acceleration [166, 156]. A simple way to include such an acceleration is to switch parameters (e.g., the maximum assimilation rate) at a certain point in the life cycle (e.g., [137, 135]). An elegant solution to accelerate development in a smooth manner is to assume a deviation from isomorphy after birth [17, 166]. In an isomorphic animal, the surface areas for food uptake and for reserve mobilisation scale with volume to the two-thirds (or inversely with a length measure). In a V1 morph, surface area scales with volume to the power one; think for example of an organism that increases in length but not in width (see Fig. 2.3). Such organisms have lost their size control (both food uptake and maintenance costs grow with body volume in the same manner) and grow exponentially.²⁴ The effect of a V1 stage is that the DEB parameters for food uptake and reserve mobilisation increase in a smooth manner. After a certain maturity threshold, there is a ‘metamorphosis’ into an isomorphic organism again. Quite a number of species have a major metamorphosis in their life cycle (discussed later in this section), which could be a candidate point for such a shift from V1-morphy to isomorphy. At this moment, the existence of a V1 stage in the life cycle remains speculative. It solves some of the problems in fitting DEB models for a number of species but whether the underlying mechanism is correct still needs to be tested in detail (e.g., by detailed measurement of the feeding rate over ontogeny).

The V1 stage leads to ‘upcurving’ in the growth curve; body length versus time becomes S-shaped instead of the expected Von Bertalanffy curve. This pattern is observed in some species, but that does not proof that the V1 stage is an appropriate assumption. S-shaped growth curves may also reflect changes in other DEB parameters with age. A particularly interesting candidate is the food supply. Early juveniles often have different food requirements than adults of the same species. Supplying an optimal food source for adults in a laboratory setting may thus, inadvertently, lead to food limitation in early juveniles [263, 114]. This issue is discussed in more detail in Section 6.3.

Plants and other autotrophs

In this book, I focus on animals, so autotrophic organisms are outside of the scope. It is however good to realise what makes these organisms different in DEB terms. Animals obtain their energy and building blocks from the same flux (feeding). Energy and building blocks therefore always arrive together, which means that we can depart from a situation with a single reserve (or even no reserve). Autotrophs obtain their energy from a different source (e.g., sunlight) than their building blocks (e.g., carbon dioxide from the air and ammonia from soil). This means that energy and building blocks do not arrive together and that one of the two can limit growth. It will thus often be necessary to delineate (at least) two reserve compartments, and we have to think about

²⁴Populations of micro-organisms can often be treated as a single V1 super organism. Individuals might grow more or less isomorphically, but divide at a certain point. When the population volume grows by a factor of two, this roughly means twice as many individuals, and thus twice as much surface area.

stoichiometry (how much of each of these two reserves is needed to build structure). For terrestrial plants, it also makes sense to think about two structures, as the ratio between root and shoot volume depends on the environmental conditions. Models with multiple structures and multiple reserves will inevitably have a lot of parameters, and thus require extensive data sets to parametrise. For more information on applying DEB with terrestrial plants, I like to point you to the overview paper of Russo *et al.* [231]. For algae and duckweed, simple TKTD approaches (basically DEB-free) have been presented that can be used as starting points for further study [164, 234].

Birds, mammals, and the supply-demand spectrum

From a metabolic perspective, species can be classified by their position in the demand-supply spectrum [175]. The ectotherms that I focus on in this book are mainly at the supply end of the spectrum: their life histories are closely linked to the food availability. Growth and reproduction rates respond rapidly to changes in food level, and many species can even shrink considerably under starvation. In demand systems, in contrast, growth and reproduction rates are largely preprogrammed and the organism has to work hard to obtain enough resources to meet this demand. Birds and mammals are clearly at the demand end of the spectrum. These animals do not have the flexibility for large changes in their growth and reproduction patterns; when food is insufficient they simply die. Growth in most endotherms is also well described by the von Bertalanffy curve, suggesting that there are close similarities to the supply-animals, in terms of metabolic organisation. Nevertheless, model adaptations will be needed to cover the energetic consequences concerning regulation of feeding and reproduction.

Reproduction in birds and mammals is not so much constrained by energetics of offspring production but by the (expected) demands of parental care. For birds and mammals, it would be unrealistic to assume a continuous allocation to reproduction after puberty. Since most birds will only breed in a specific part of the year, continuous allocation would imply that birds save up resources for most of the year to produce a clutch of eggs.²⁵ Furthermore, such continuous allocation would be troublesome to reconcile with the fact that birds are able to replace an egg if one is removed from the nest, and the fact that some birds are able to produce a second (or even third) brood in the same season, but only if food conditions allow. Furthermore, if a bird cannot find a mate in one year, it will not increase in weight, nor produce twice as many eggs in the subsequent year. Similarly, humans that do not reproduce do not increase in weight due to the continuous build up of a reproduction buffer.

The answer to this conundrum lies in up-regulation: we see that feeding is up-regulated during egg production (birds) or pregnancy/nursing (mammals). Such up-regulation is a tricky process in the DEB context. If a standard DEB animal increases its feeding rate, it should grow larger in structure, which is not what we see in birds and mammals. A logical option to modify the model would be to redirect the additional resources obtained from up-regulation completely to reproduction, bypassing the reserve, and thereby also the κ split (thus leaving the somatic branch unaffected). For birds, a modification of the standard DEB animal model was presented by Marn

²⁵This would result in an estimated value for κ that is very close to 1, which was indeed produced by DEB analyses for various birds: an average κ of 0.988 across 40 species [246]. In a more detailed analysis for wandering albatross [247], a κ value of 0.997-0.999 was reported. Such values seem unrealistic, and point at a problem with the model's structure.

et al. [191] in which up-regulation of feeding is considered, through two hypotheses (Fig. 2.7). Interestingly, the authors found that the most likely pathway was for the additional resources to go into the reserve. To avoid an increase in structural body size, somatic maintenance would be increased in accordance with the increase in feeding rate. In the model testing by the authors, this hypothesis outperformed the alternative of up-regulation going straight into reproduction. The modified DEB model, as proposed by Marn *et al.*, still sports continuous allocation to reproduction, also outside of the breeding season (albeit with a smaller flux into the reproduction buffer, owing to a lower feeding rate). The model was tested with data on reared bobwhite quail, kept under the conditions for toxicity testing. The test protocol prescribes daily removal of the produced eggs, such that the females will indeed have an almost constant (and unnaturally high) reproduction rate. Under such conditions, continuous allocation to reproduction may be reasonable. However, I would like to see how this model holds up for birds under more natural conditions. In a more recent publication, Trijau *et al.* [249] select the alternative hypothesis (additional resources go directly into the reproduction buffer) for their model for birds.

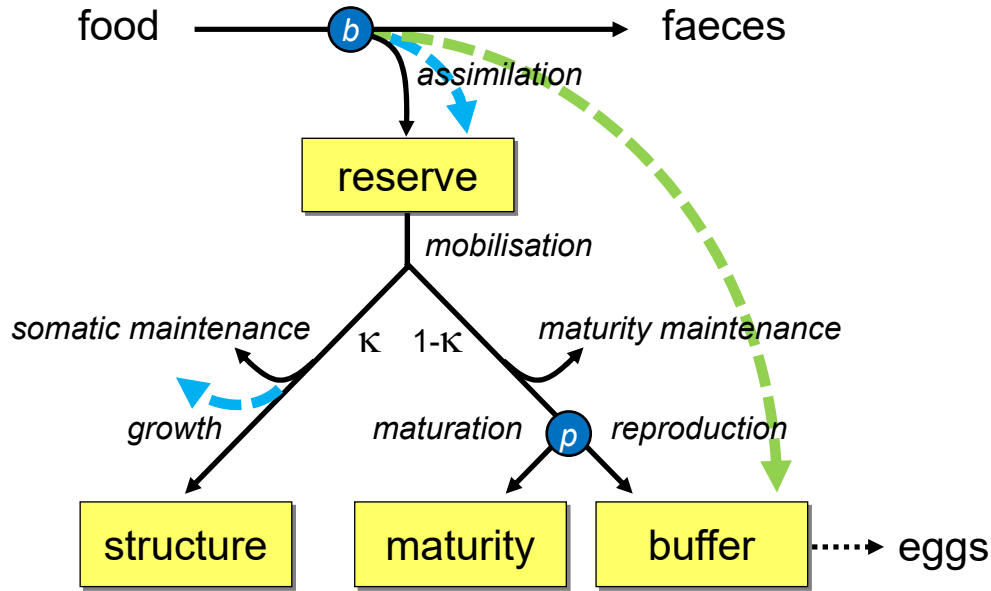


Figure 2.7: Schematic diagram of the energy flows in a standard DEB animal, showing the alternative hypotheses for birds. Related to egg production, feeding is up-regulated. These extra resources could be used directly for reproduction (green arrow), or go into the reserve with an additional cost for somatic maintenance (blue arrows). Scheme modified from [191].

The model adaptations needed for endotherms was one reason to exclude them from a thorough discussion in this book. Another reason is that toxicology for these organisms generally focusses on the health and well-being of the individual, rather than on population-relevant energy-requiring traits such as growth and reproduction. As an extreme example, it would not be very interesting to know when a certain chemical affects the energy available for growth and reproduction in humans. More subtle changes at the sub-individual level would be the endpoint of choice (such as enzyme or tumour induction, or malformations in the embryo). We can still use TKTD models for such endpoints, but the TD part would not generally be an energy-budget

approach. There may, however, be toxicological applications where DEB models can provide useful insights. Interesting first studies into using a simplified DEBkiss model for toxicology in mammals were published for mink (*Mustela vison*) [57] and rat (*Rattus norvegicus*) [199, 198]. These authors, however, did not modify the model to deal with up-regulation of feeding during pregnancy and nursing (and focussed mostly on growth).

Reproductive modes

The standard life cycle of Figure 2.4 suggest a parthenogenetic, egg-producing animal. In such species, offspring are produced asexually by the mother (although many of these species are also capable of sexual reproduction). Examples include most water fleas (Cladocera), stick insects, some nematodes (e.g., *Acrobelloides nanus*), and the popular test species in soil ecotoxicology, the springtail *Folsomia candida*. This does not mean, however, that DEB theory is limited to such life cycles. It does mean that we have to think a little bit when dealing with other species, and, if needed, modify the standard model to some extent. A thorough treatment of reproduction behaviour is outside of the scope of this book, but the overview below should give you a better idea of what is going on.

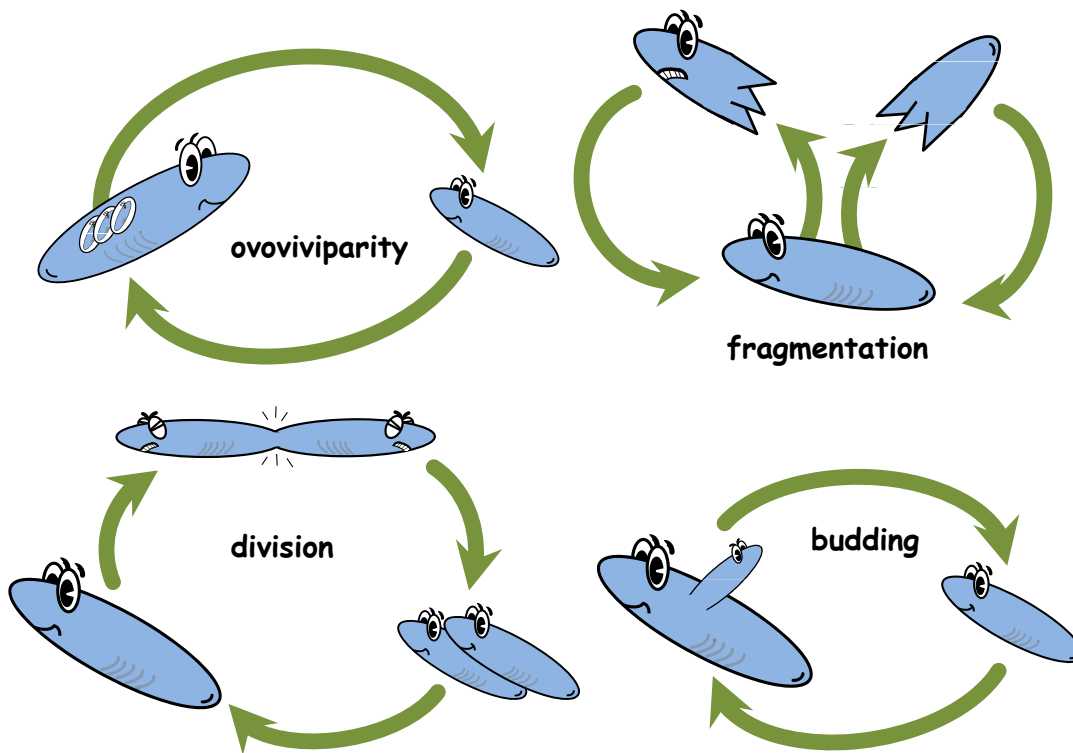


Figure 2.8: Examples of asexual reproduction without (external) eggs.

Ovoviviparity. Some parthenogenetic species do produce eggs, but hatch them internally, so that life offspring are introduced into the outside world (Fig. 2.8).²⁶ Examples

²⁶Sexually reproducing organisms may also be ovoviviparous (such as guppies), which requires internal fertilisation.

are aphids and the water fleas (including the popular test species *D. magna*). In essence, this does not make a big difference from the standard life cycle, apart from the fact that the embryonic phase happens inside the mother, which raises a question of timing. Take *Daphnia* as an example: after moulting, the female places eggs in her brood pouch, where they develop, hatch, and are released at the next moult. The experimenter now counts the offspring when they emerge from the mother, and calls that reproduction at time t . From an energetic viewpoint, however, reproduction or spawning is the formation of the eggs, which happens several days earlier [214]. When comparing reported reproduction to the model predictions, this time shift needs to be considered to construct a consistent energy budget [142, 140].

Other asexual modes of reproduction. Some organisms can reproduce asexually by division, fragmentation or budding (Fig. 2.8). When they do not produce eggs or embryos at all, they are considered ‘eternal juveniles’ in the DEB context. Dividing animals include amoeba and ciliates, fragmentation occurs in some annelid worms and sea stars, and hydra is a well-known budding animal. This mode of reproduction requires some attention in the theory, especially when the division is asymmetric. Populations of dividing organisms may be treated as a single V1-morphic organism (consult [155] for more details).

Sexual reproduction: hermaphrodites. In hermaphrodites, the animals combine the male and female function into a single individual. Some need to mate, but others can also self-fertilise. Popular test species include the nematode *Caenorhabditis elegans* and the great pond snail *Lymnaea stagnalis*. In principle, the same life cycle picture applies as for parthenogenetic species. However, not all of the mobilised reserve allocated to reproduction will be used for eggs; a part will be used to produce sperm. The contribution of sperm (and other aspects of the male sexual function) on the energy budget is not as easily quantified as egg production. However, experiments with surgically altered pond snails (to stop the male function) indicate that, in these hermaphrodites, the male function is roughly taking the same amount of energy as the flux to the eggs [56].

The nematode *C. elegans* is a particularly interesting case, as it is a sequential hermaphrodite. It first produces a complement of sperm cells and, at some point, sperm production halts, and the animal switches to egg production [259]. The eggs are fertilised in the hermaphrodite with its own stored sperm cells (unless a male is present). Interestingly, the number of sperm cells is limited in *C. elegans*: roughly 300 are produced, which are used to fertilise the same number of eggs. When the sperm storage runs out, egg production stops too (unless males are present to provide fresh sperm cells). It seems that, under stress, sperm production can be reduced to move the production of the first eggs forward, at the expense of total reproductive output [1]. The proper way to address this mode of reproduction in DEB would likely be to include two maturity thresholds for puberty (start of sperm production and switch to egg production), and two reproduction buffers (for sperm and eggs). In practice, more ‘quick-and-dirty’ solutions have been used, so far [114, 81].

Sexual reproduction: males and females. In ecotoxicity testing, the use of sexually-reproducing species is rather rare, and when such species are used, the focus lies on the

females. This is to some extent defensible from the point of view of the population. In many species, a single male suffices to fertilise a large number of females, which implies that the health of the females has more influence on (short-term) population performance than that of males. Males often have different properties than females (e.g., size, growth rate, reproductive investment), which means they differ in their DEB parameters. In fact, males and females can be treated as different species (although hopefully most parameters are similar). The reproductive investment for the males is more difficult to quantify than for the females, which complicates the derivation of a full parameter set for males.

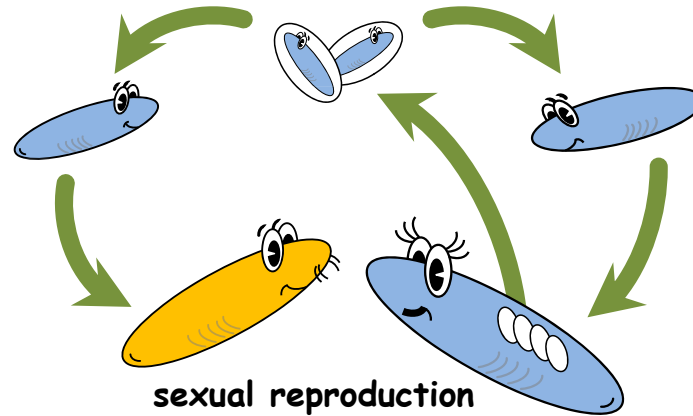


Figure 2.9: Sexual reproduction with males and females.

Metamorphosis

Many species sport a large transformation at some point of their life cycle; some examples are shown in Figure 2.10. Development in DEB theory is tightly linked to the concept of maturity. It thus makes sense to use maturity thresholds to trigger metamorphosis and to demarcate the stages in development [17, 203]. Parameters may change with metamorphosis, but whether they always do is an open question. Hopefully, most parameters keep their value throughout the life cycle, otherwise, parametrising a DEB model will become a difficult task.

Amphibians have been modelled with DEB ([203] and several entries in the add-my-pet library, see [192]). A detailed DEB model for the anuran (frogs and toads) metamorphosis was presented more recently by Romoli *et al.* [226]. The interested reader may consult these papers.²⁷

For holometabolic insects, it seems plausible that the metamorphosis is accompanied by different parameter values as the adult is both morphologically and physiologically extremely different from the larvae. Furthermore, it is likely that there are more structural differences in the metabolic organisation as these animals do not reach their maximum size asymptotically, as predicted from DEB theory (see next section). Instead, they emerge from the pupae as an adult that does not grow anymore. Some detailed work on insects in a DEB context has already been performed [179, 186, 152, 148]. However, these models are all different, and there is no consensus on the ‘correct’ one.

²⁷I must admit that I have not (yet) studied these papers, also because these species fall outside of the main scope of this book (see Section 1.1).

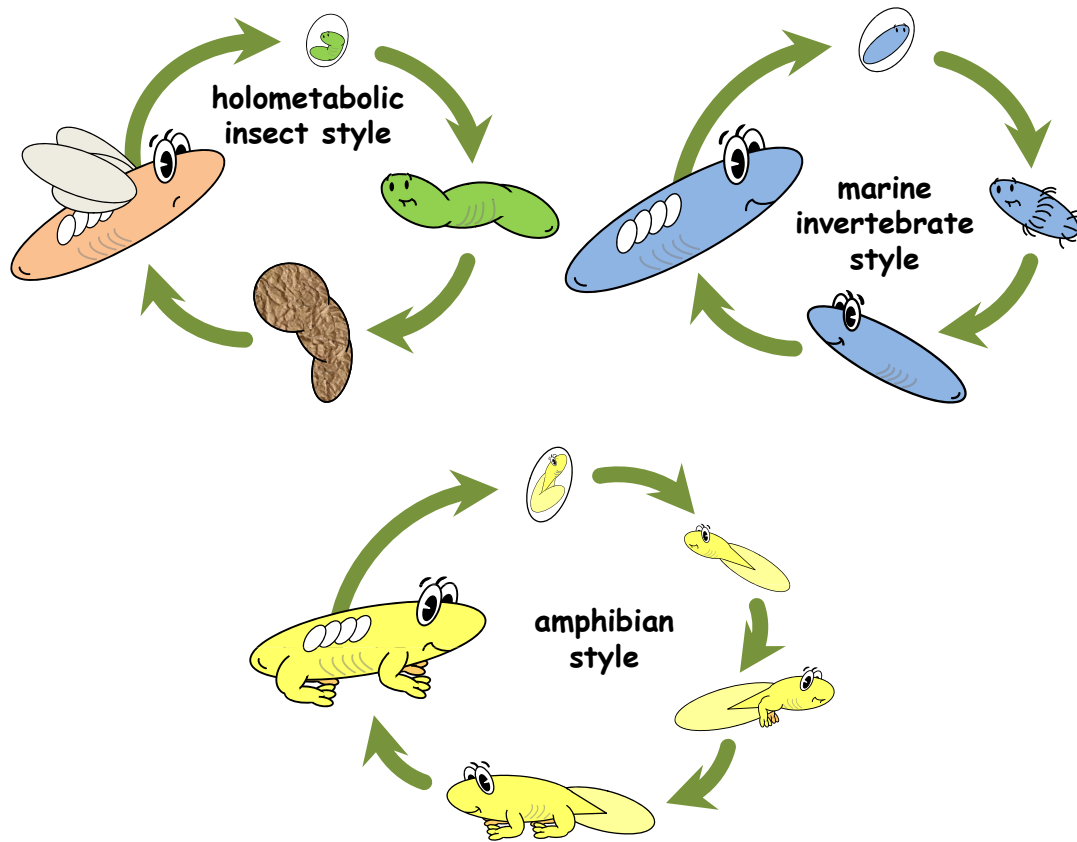


Figure 2.10: Examples of metamorphosis in animal life cycles. Holometabolic insects (e.g., butterflies and beetles) have clearly different larval and adult forms, with a pupal stage in between. Many marine invertebrates have a larval stage that lives in the plankton. Such a life cycle can be found in echinoderms (a.o., starfish and sea urchins), marine molluscs (a.o., marine slugs and bivalves) and marine polychaetes (annelid worms). Most amphibians hatch from the egg as a tadpole, which gradually transforms into the adult form.

Further, the modified DEB models for holometabolic insects currently do not seem to provide a realistic response to food limitation during the larval stages [75] (at least for the midge *Chironomus riparius*, which hardly eats as an adult). Thus, more work is needed to provide a satisfactory inclusion of these life cycles into the theory (with an evolutionary rationale for how to developed to be so different from other animals).

Metamorphosis may be accompanied by substantial changes in shape. This is obvious in holometabolic insects; a caterpillar looks nothing like the butterfly that it will develop into. Also, many marine invertebrates have swimming (or floating) larval stages, intended for dispersal, that look nothing like the adult. This implies that we have to be careful to use the physical length of the organism in our DEB modelling efforts. The relationship between physical length and volumetric length (cubic root of volume) may be very different in the various life stages, requiring a close look. Some examples for copepods illustrate how this can be done [138, 128].

Determinate growth

All winged insects (Pterygota) have a final moult to adulthood in their life cycle, and they do not grow as an adult. Some have a metamorphosis to the adult form (holometabolism, discussed above), while others develop in more gradual steps to the adult form (hemimetabolism). Since growth abruptly ceases after the final moult, we cannot use the κ rule for allocation in the adult stage (see [179, 147]). The growth pattern is also different from the typical von Bertalanffy curve: upcurving over most of the larval stage, and a rather abrupt attainment of maximum size, instead of a smooth asymptotic approach.

Interestingly, copepods have a very similar life history as insects: they go through a fixed number of development stages and stop growth abruptly after their final moult to adulthood (although the growth curve does not tend to show smooth upcurving). Copepods have been treated in the framework of a reserve-less DEBkiss model [138, 137], but important open questions remain. It is conceivable that insects and copepods require a similar modification of the standard DEB animal model. However, in an evolutionary sense, these two groups are not very closely related; it is likely that both groups developed this lifestyle independently. As mentioned in the previous section, several DEB models for insects have been presented, but this is still very much work in progress.

Nematodes also have a fixed number of moults, with a final moult to the adult stage. However, they do show growth within each stage (the outer skin is, to some extent, flexible and allows growth, also in the final, adult, stage). These animals do grow according to the von Bertalanffy curve, albeit with initial slower growth [114]. At this moment, it is however unclear to me whether the maximum size of a nematode is governed by a balance between assimilation and maintenance needs (as predicted by DEB theory), or by the physical limitations of the adult skin (which would bring them more in line with the insects, in terms of required DEB model adaptations).

Since insects, copepods and nematodes constitute an important part of the invertebrate world (in terms of numbers of species, numbers of individuals, biomass, and ecological importance), further study on the energetics of their life histories is urgently needed. More specifically, since insects form the majority of the described species of animal,²⁸ it is disconcerting that we do not have a convincing DEB model for this group yet.²⁹ Bottleneck is, in my opinion, the lack of detailed experimental work, aimed at clarifying the energy and mass fluxes (and the switches in them at various life stages).

²⁸Of the 1.7 million species that are described by science, 1 million are insects. See <https://currentresults.com/Environment-Facts/Plants-Animals/number-species.php>, accessed 20 June 2025.

²⁹This is particularly true for holometabolic insects. For at least some hemimetabolic species, the model adaptations may be more straightforward [147]. Interestingly, the springtails (hexapods closely related to the insects) seem to follow the standard DEB models more closely (e.g., [121, 87, 125]).